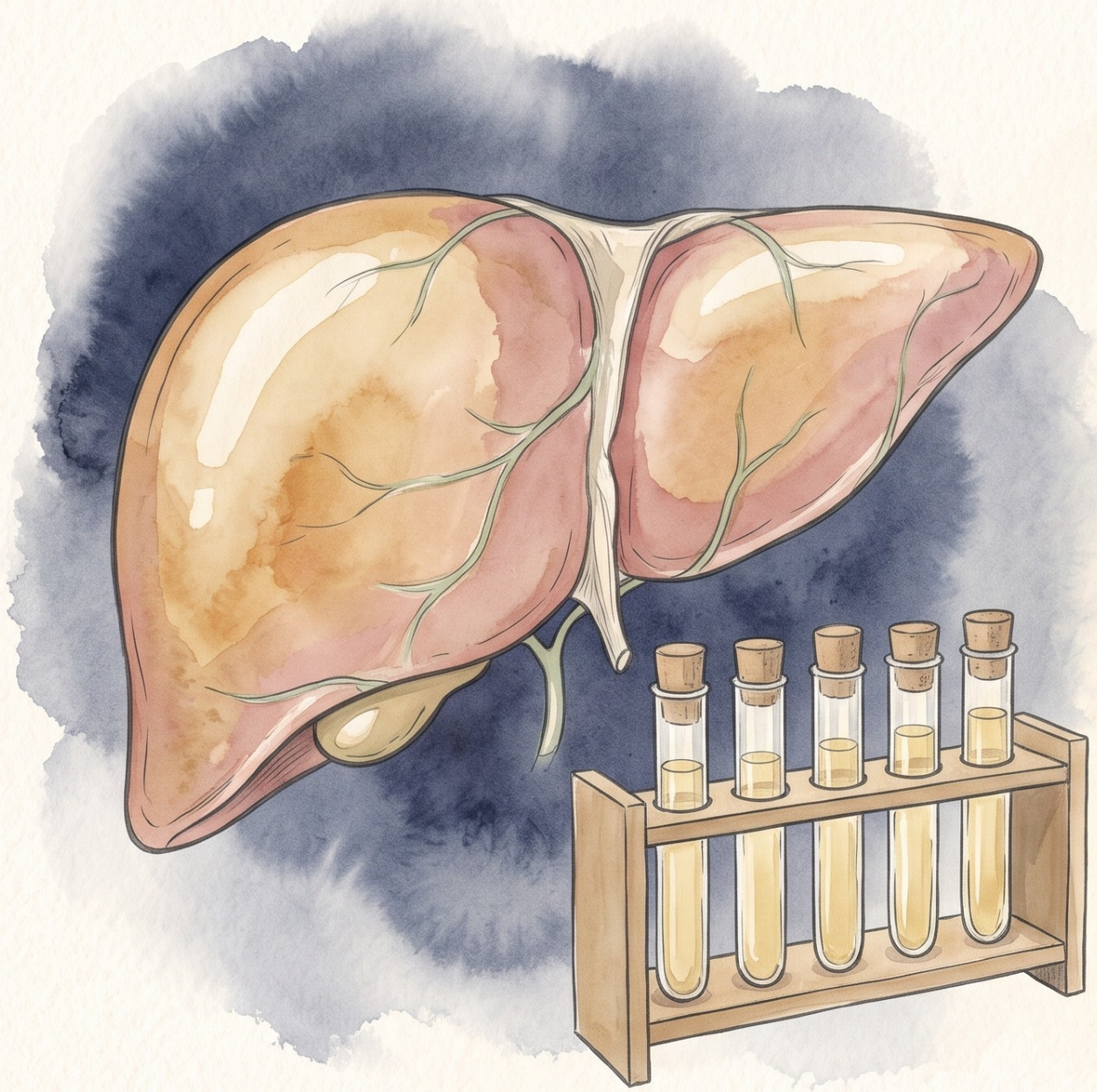


Chronic Hepatitis B

Mohammad Almeqdadi, MD
Transplant Hepatologist and Gastroenterologist
Assistant Professor of Medicine and Surgery
X: @mfalmeqdadi

Tufts UNIVERSITY | School of Medicine

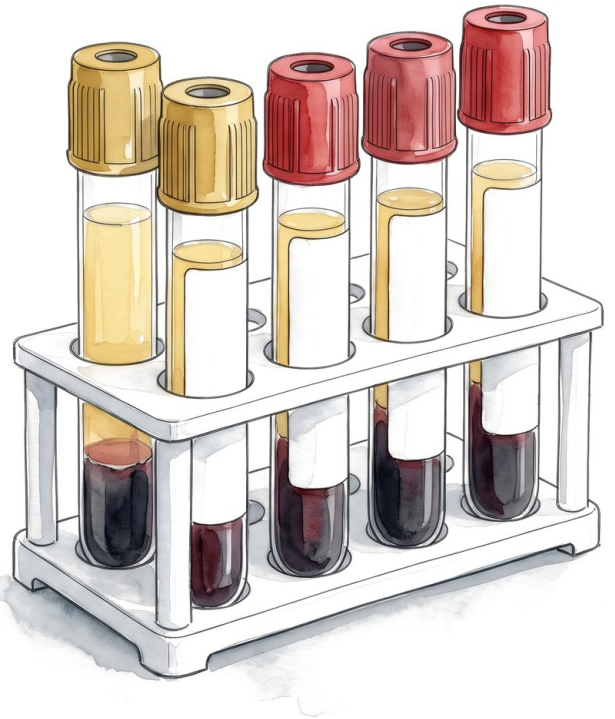
Slides (PDF)

Illustrations generated with Google Gemini

Objectives

By the end of this talk you should be able to:

- Read a hepatitis B serology panel
- Place a patient in the right phase
- Decide who to treat, using the current guidelines
- Estimate HCC risk and set up surveillance
- Prevent reactivation and transmission to the baby



The Australia antigen

A "New" Antigen in Leukemia Sera

The "Australia antigen" is found in the sera of some normal individuals from foreign populations. The total absence of the antigen from the sera of normal United States subjects and its relatively high frequency in acute leukemia suggests that the presence of the antigen may be of value in the diagnosis of early acute leukemia. Whether the antigen results from or precedes the leukemia process remains to be seen.

In 1963, sera from patients with hemophilia who had received transfusions were tested for the presence of isoprecipitins using a panel of 24 sera from normal individuals, including sera from foreign populations. Two of the hemophilia sera formed a clearly defined precipitin line with one of the panel sera (from an Australian aborigine), but with none

- Found in the serum of some normal people from foreign populations, and in none of the normal US subjects
- Frequent in acute leukemia, which is why the paper is about leukemia
- Blumberg shared the 1976 Nobel Prize with Carleton Gajdusek

Blumberg BS, Alter HJ, Visnich S. JAMA 1965;191:541-546. PMID 14239025.

From antigen to vaccine

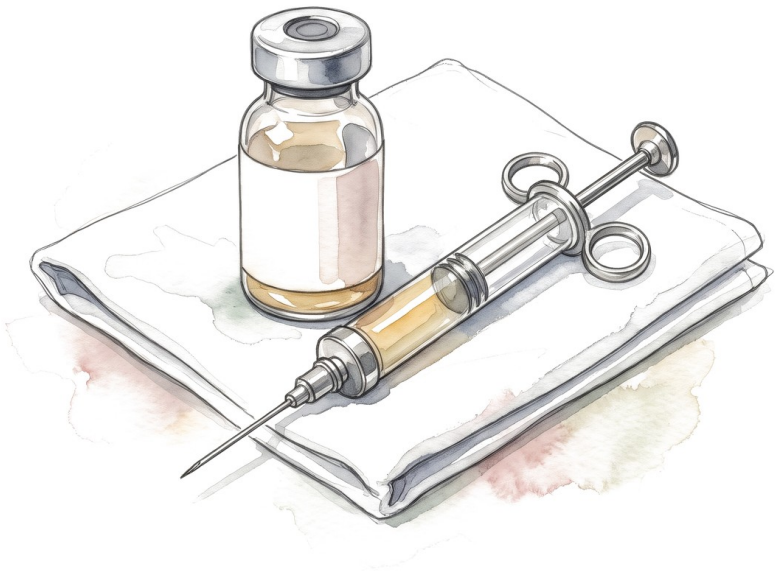
1965 — Blumberg, Alter and Visnich describe the Australia antigen

1970 — Dane, Cameron and Briggs see the virus particle in serum

1976 — Blumberg and Gajdusek receive the Nobel Prize. First immunization of people with plasma-derived vaccine (Maupas)

1980 — Szmuness: the vaccine works in a controlled trial of 1,083 men

1981 — Beasley: HBsAg carriers in Taiwan get liver cancer at a far higher rate



PMIDs: Dane 4190997; Maupas 59012; Szmuness 6997738; Beasley 6118576.

Prince, 1968

*AN ANTIGEN DETECTED IN THE BLOOD DURING THE INCUBATION PERIOD OF SERUM HEPATITIS**

BY ALFRED M. PRINCE

LABORATORY OF VIROLOGY, THE NEW YORK BLOOD CENTER, AND DEPARTMENT OF PATHOLOGY, THE NEW YORK HOSPITAL-CORNELL MEDICAL CENTER, NEW YORK CITY

Communicated by Henry G. Kunkel, April 8, 1968

The aim of the following experiments was to provide an objective immunologic criterion for the diagnosis of serum hepatitis, as well as a possible means of screening for carriers of the agent of this disease. An antigen that reacted in the immunodiffusion test with serum from multiply transfused patients was detected in the blood during the incubation period prior to the onset of major chemical or

- Okochi and Murakami reported the antigen in Japanese the same year

Dane, 1970

VIRUS-LIKE PARTICLES IN SERUM OF PATIENTS WITH AUSTRALIA-ANTIGEN-ASSOCIATED HEPATITIS

D. S. DANE C. H. CAMERON
MOYA BRIGGS
Bland-Sutton Institute, Middlesex Hospital, London W.1

Summary Virus-like particles about 42 nm. in diameter have been found in multiple serum specimens from three Australia-antigen-positive hepatitis patients. It is suggested that these particles may be complete virus and that the much more numerous 22 nm. particles and long forms of Australia antigen are surplus virus-coat material.

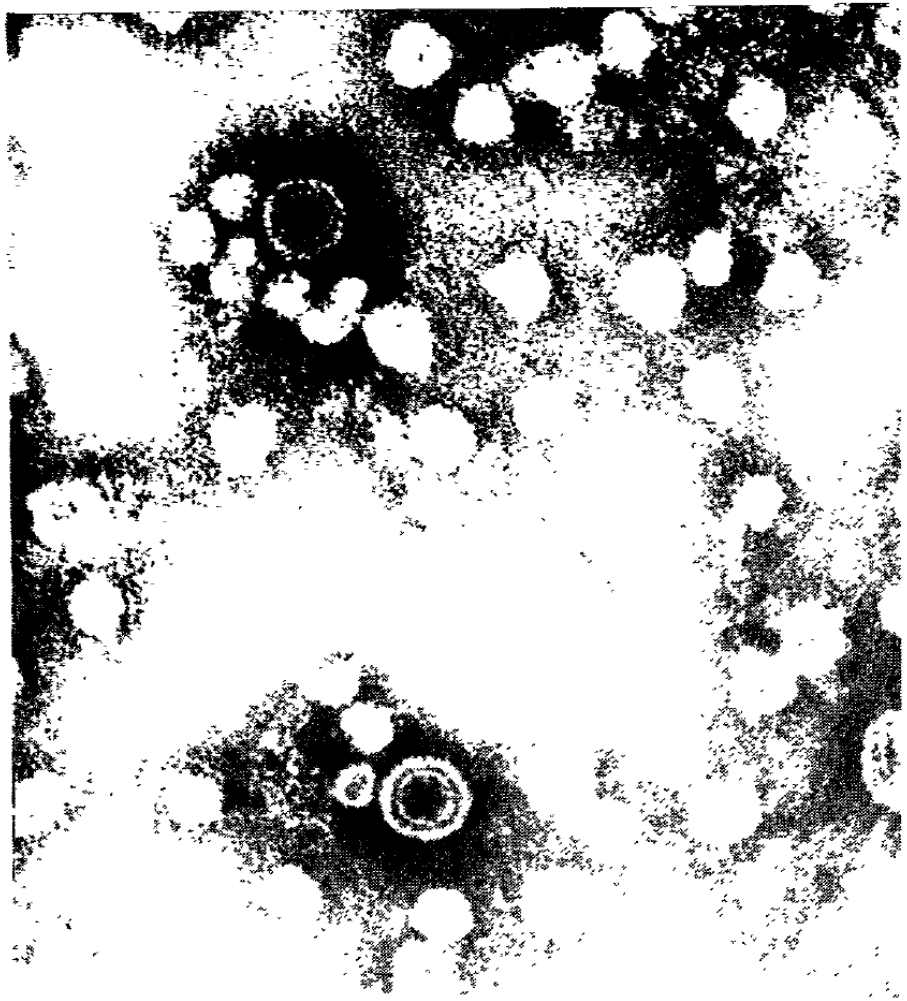


Fig. 1.— Gradient-purified particles from the serum of patient 2.

Dane DS, Cameron CH, Briggs M. Lancet 1970;1:695-698. PMID 4190997.

Beasley, Taiwan, 1981

HEPATOCELLULAR CARCINOMA AND HEPATITIS B VIRUS

A Prospective Study of 22 707 Men in Taiwan

R. PALMER BEASLEY LU-YU HWANG
CHIA-CHIN LIN CHIA-SIANG CHIEN

University of Washington Medical Research Unit, Taipei; Institute of Public Health, College of Medicine, National Taiwan University; and Government Employees' Clinic Centre, Taipei, Taiwan

Summary A prospective general population study of 22 707 Chinese men in Taiwan has shown that the incidence of primary hepatocellular carcinoma (PHC) among carriers of hepatitis B surface antigen (HBsAg) is much higher than among non-carriers (1158/100 000 *vs* 5/100 000 during 75 000 man-years of follow-up). The relative risk is 223. PHC and cirrhosis accounted for 54·3% of the 105 deaths among HBsAg carriers but accounted for only 1·5% of the 202 deaths among non-carriers. These findings support the hypothesis that hepatitis B virus has a

	HBsAg carriers	Non-carriers
HCC incidence per 100,000	1,158	5
Deaths from HCC or cirrhosis	54.3% of 105	1.5% of 202

Relative risk 223

Beasley RP et al. Lancet 1981;2:1129-1133. PMID 6118576.

Szmuness, 1980

Volume 303 OCTOBER 9, 1980 Number 15

HEPATITIS B VACCINE

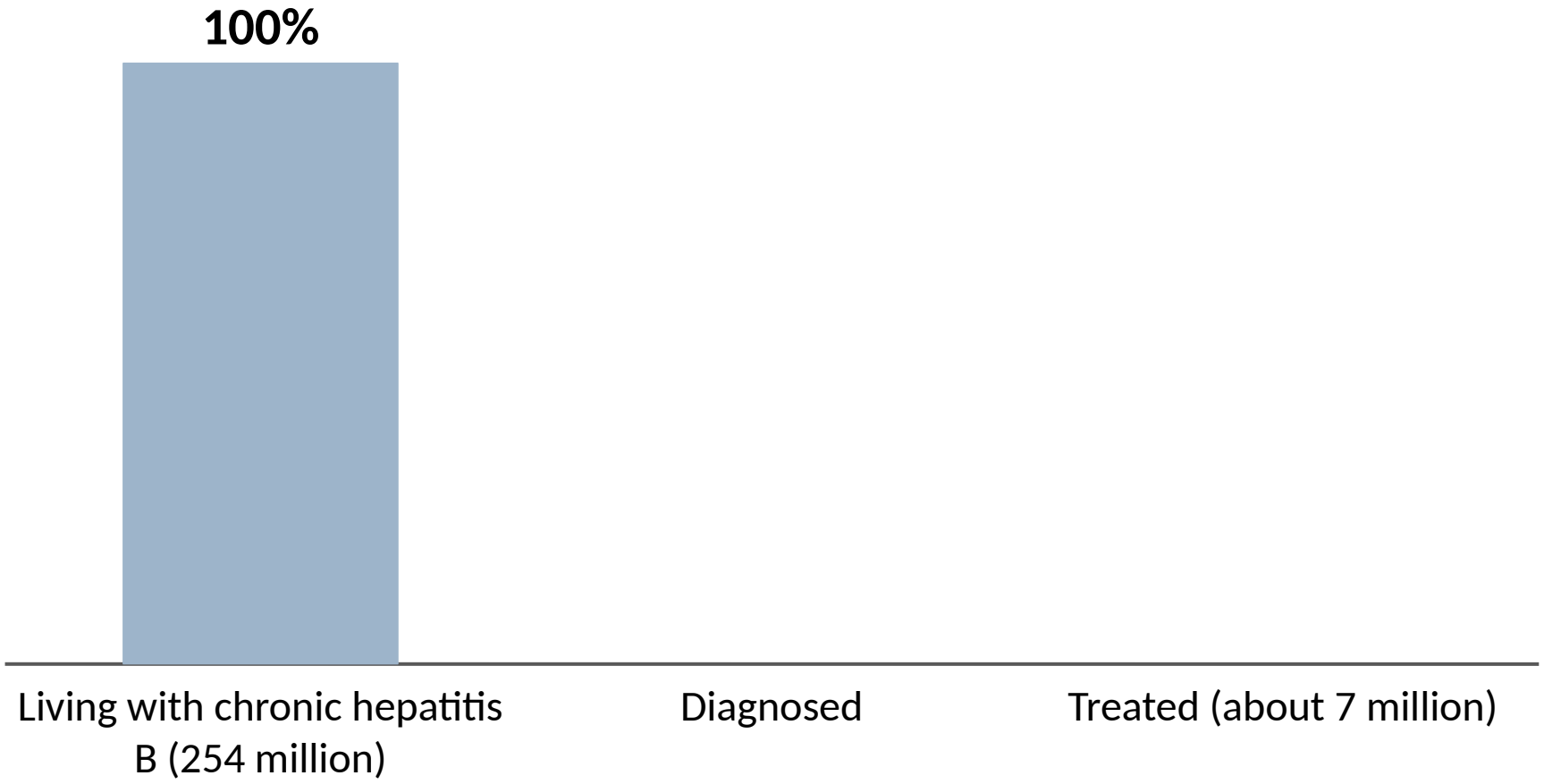
Demonstration of Efficacy in a Controlled Clinical Trial in a High-Risk Population in the United States

WOLF SZMUNESS, M.D., CLADD E. STEVENS, M.D., EDWARD J. HARLEY, M.S., EDITH A. ZANG, PH.D., WILLIAM R. OLESZKO, PH.D., DANIEL C. WILLIAM, M.D., RICHARD SADOVSKY, M.D., JOHN M. MORRISON, AND AARON KELLNER, M.D.

Abstract We assessed the efficacy of an inactivated hepatitis B vaccine in a placebo-controlled, randomized, double-blind trial in 1083 homosexual men known to be at high risk for hepatitis B virus infection. The vaccine was found to be safe, and the incidence of side effects was low. Within two months, 77 per cent of the vaccinated persons had high levels of antibody against the hepatitis B surface antigen. This rate increased to 96 per cent after the booster dose and remained essentially unchanged for the duration of the trial. For the first 18 months of follow-up, hepatitis B or subclinical infection developed in only 1.4 to 3.4 per cent of the vaccine recipients as compared with 18 to 27 per cent of placebo recipients ($P<0.0001$). The reduction of incidence in the vaccinees was as high as 92.3 per cent; none of the vaccinees with a detectable immune response to the vaccine had clinical hepatitis B or asymptomatic antigenemia. A significant reduction of incidence was already seen within 75 days after randomization; this observation suggests that the vaccine may be efficacious even when given after exposure. (N Engl J Med. 1980; 303:833-41.)

Hepatitis B worldwide

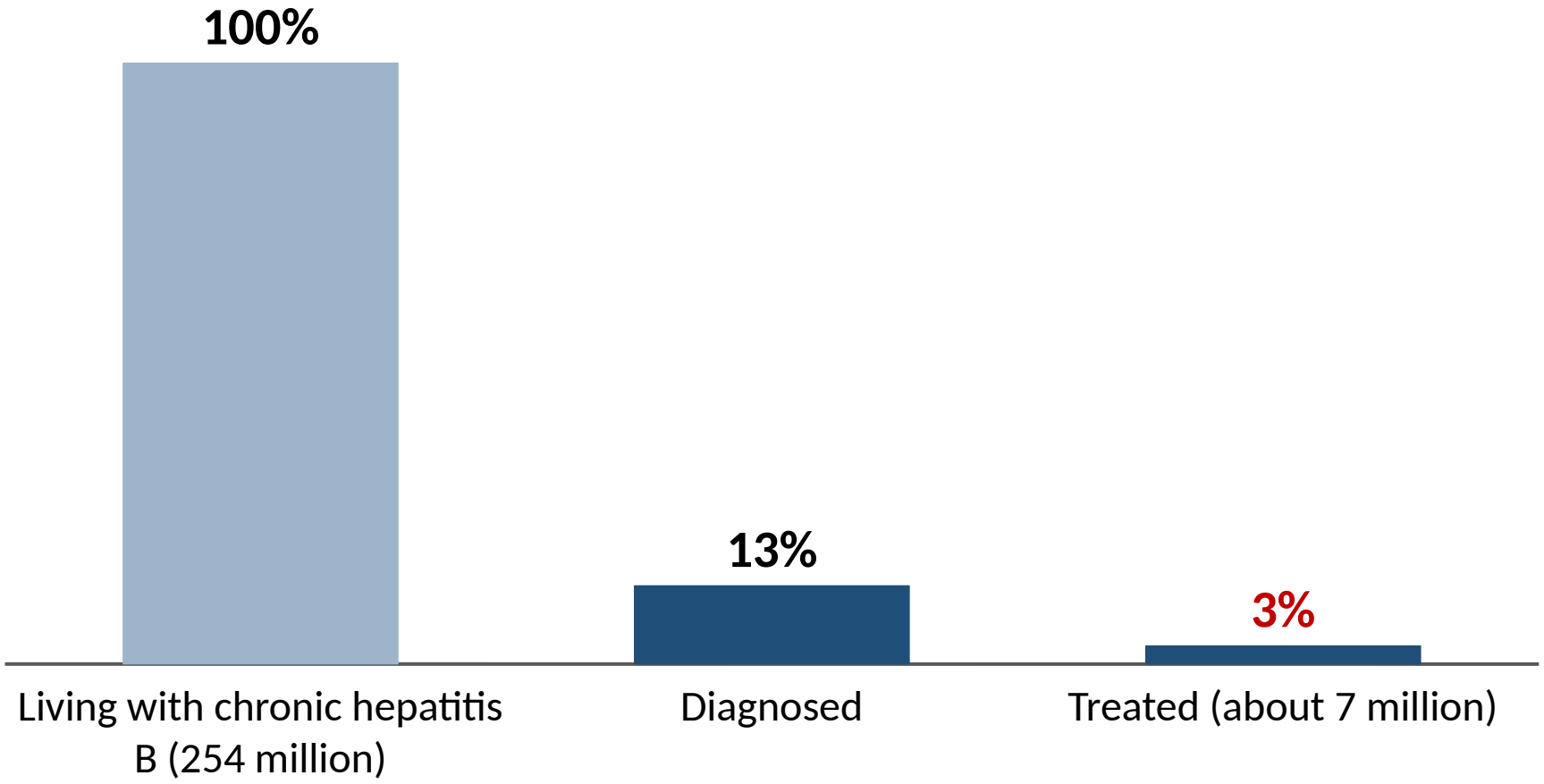
How many people with chronic hepatitis B know it?



WHO news release 9 April 2024 (2022 estimates), rounded. WHO Global hepatitis report 2024.

Hepatitis B worldwide

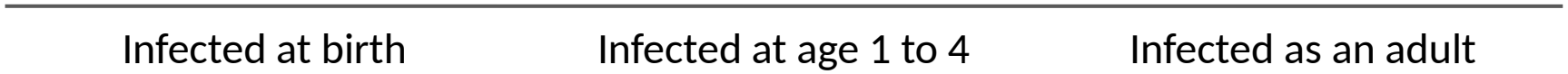
How many people with chronic hepatitis B know it?



WHO news release 9 April 2024 (2022 estimates), rounded. WHO Global hepatitis report 2024.

Who becomes chronic?

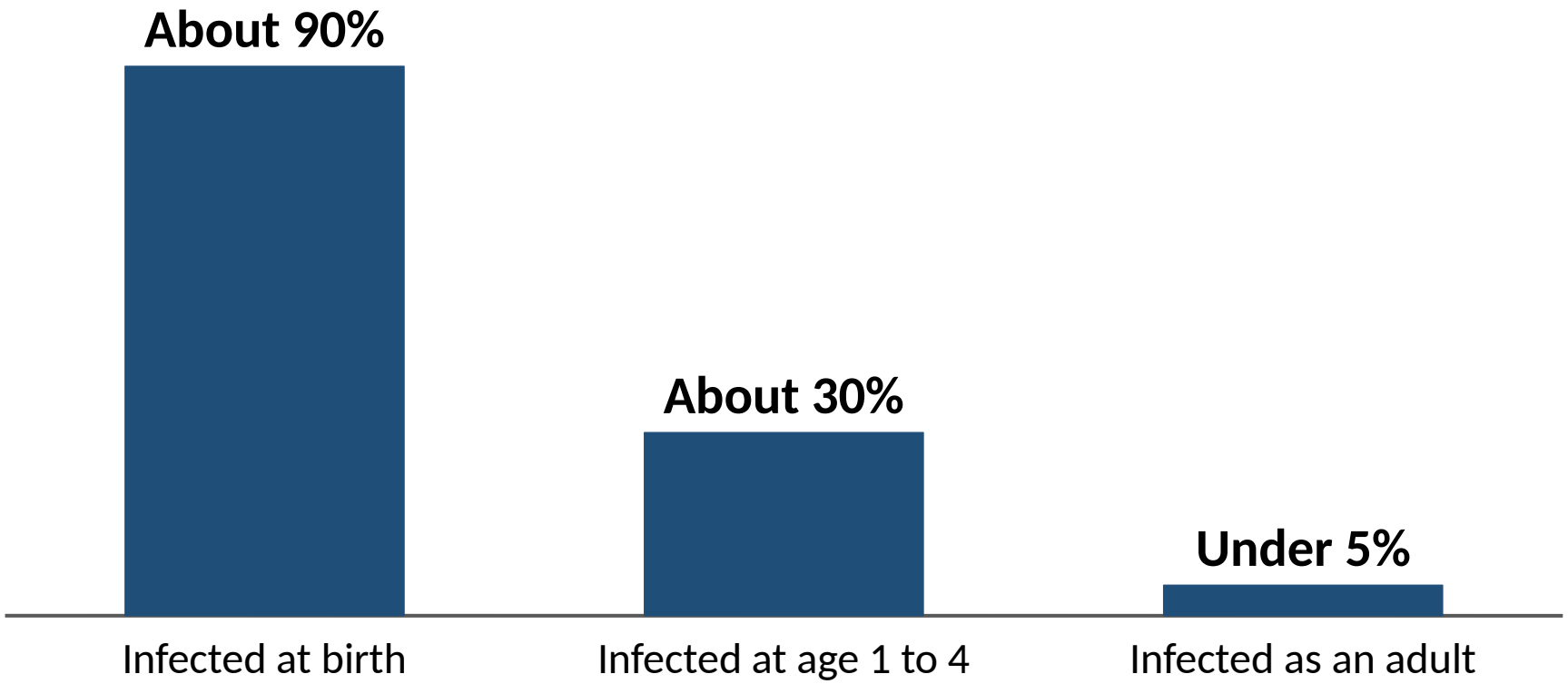
How often does infection become chronic, by age at infection?



WHO guidelines for chronic hepatitis B, 2024. CDC perinatal hepatitis B overview, 2025.

Who becomes chronic?

How often does infection become chronic, by age at infection?



WHO guidelines for chronic hepatitis B, 2024. CDC perinatal hepatitis B overview, 2025.

Hepatitis B in the United States

Estimate	People with chronic HBV	Source
NHANES 2013 to 2018	880,000 (580,000 to 1.17 million)	Household survey
Meta-analysis including immigrants	1.89 million (1.49 to 2.4 million)	1.47 million are foreign-born
Not aware of the infection	About two-thirds	CDC 2023
Born outside the US	About 69%	CDC 2023

Conners EE et al. MMWR Recomm Rep 2023;72(1):1-25. PMID 36893044.

Screening an adult for hepatitis B

Which three tests make up the screening panel?

Question	Test
Is there a current infection?	
Is the patient immune?	
Has the patient ever been exposed?	

CDC recommendations, MMWR Recomm Rep 2023;72(1):1-25, PMID 36893044.

Screening an adult for hepatitis B

Which three tests make up the screening panel?

Question	Test
Is there a current infection?	HBsAg
Is the patient immune?	Anti-HBs
Has the patient ever been exposed?	Total anti-HBc

Once for every adult 18 and older, and in every pregnancy

CDC recommendations, MMWR Recomm Rep 2023;72(1):1-25, PMID 36893044.

Five panels

What does each panel mean?

	HBsAg	Total anti-HBc	IgM anti-HBc	Anti-HBs	Interpretation
Panel 1	–	–	–	–	
Panel 2	–	–	–	+	
Panel 3	–	+	–	+	
Panel 4	+	+	–	–	
Panel 5	+	+	+	–	

Interpretation per CDC 2023 (Table 1), MMWR Recomm Rep 2023;72(1):1-25.

Five panels

What does each panel mean?

	HBsAg	Total anti-HBc	IgM anti-HBc	Anti-HBs	Interpretation
Panel 1	–	–	–	–	Susceptible
Panel 2	–	–	–	+	Vaccinated
Panel 3	–	+	–	+	Resolved infection
Panel 4	+	+	–	–	Chronic infection
Panel 5	+	+	+	–	Acute infection

Interpretation per CDC 2023 (Table 1), MMWR Recomm Rep 2023;72(1):1-25.

61-year-old woman starting rituximab

HBsAg	Total anti-HBc	Anti-HBs	HBV DNA	ALT
–	+	–	Not done	28 U/L

What could this be, and what do you order?

Possibility	How you sort it out

CDC 2023 (Table 1), PMID 36893044; AASLD 2018 hepatitis B guidance, PMID 29405329.

61-year-old woman starting rituximab

HBsAg	Total anti-HBc	Anti-HBs	HBV DNA	ALT
–	+	–	Not done	28 U/L

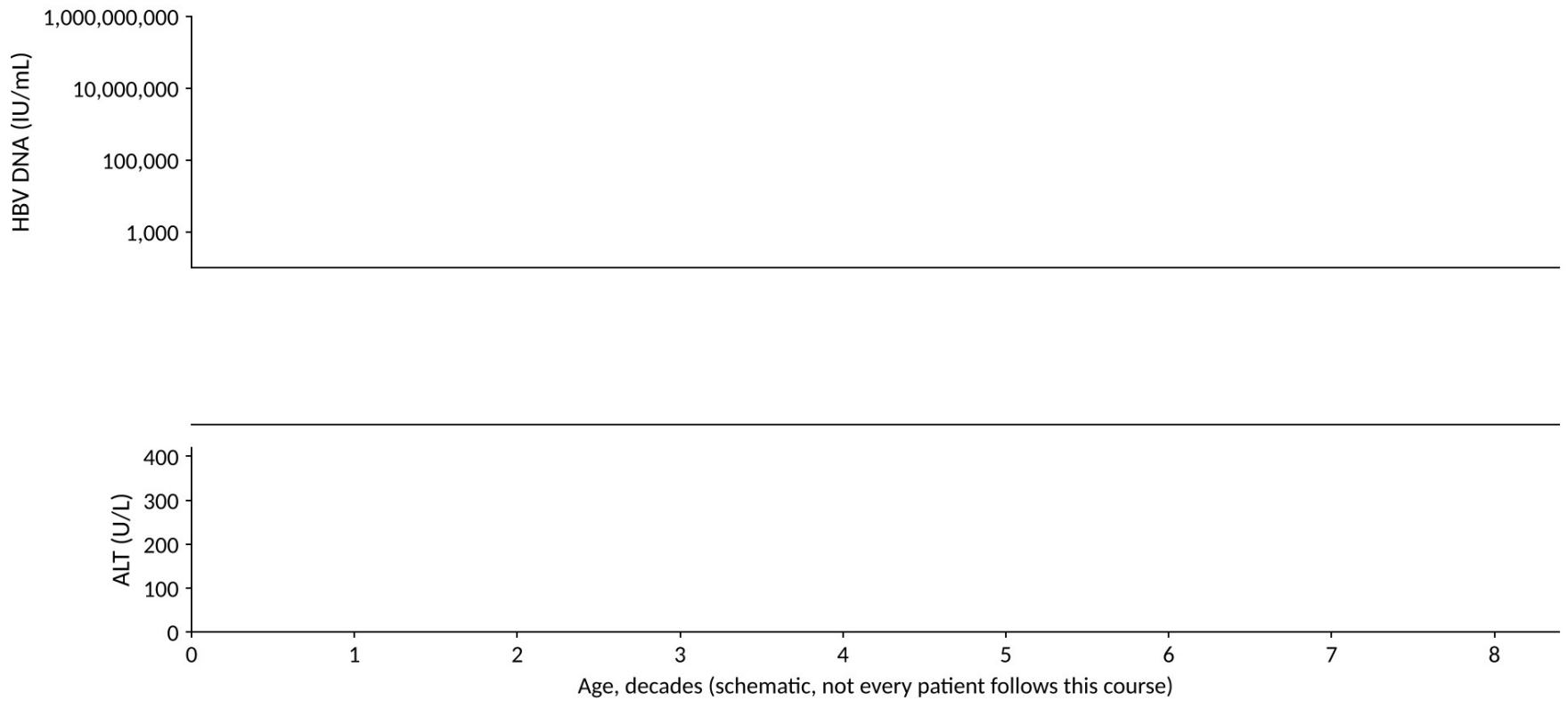
What could this be, and what do you order?

Possibility	How you sort it out
Resolved infection, anti-HBs has waned	HBV DNA undetectable
Occult infection	HBV DNA detectable
Window period	IgM anti-HBc positive
False-positive anti-HBc	Repeat testing, DNA undetectable

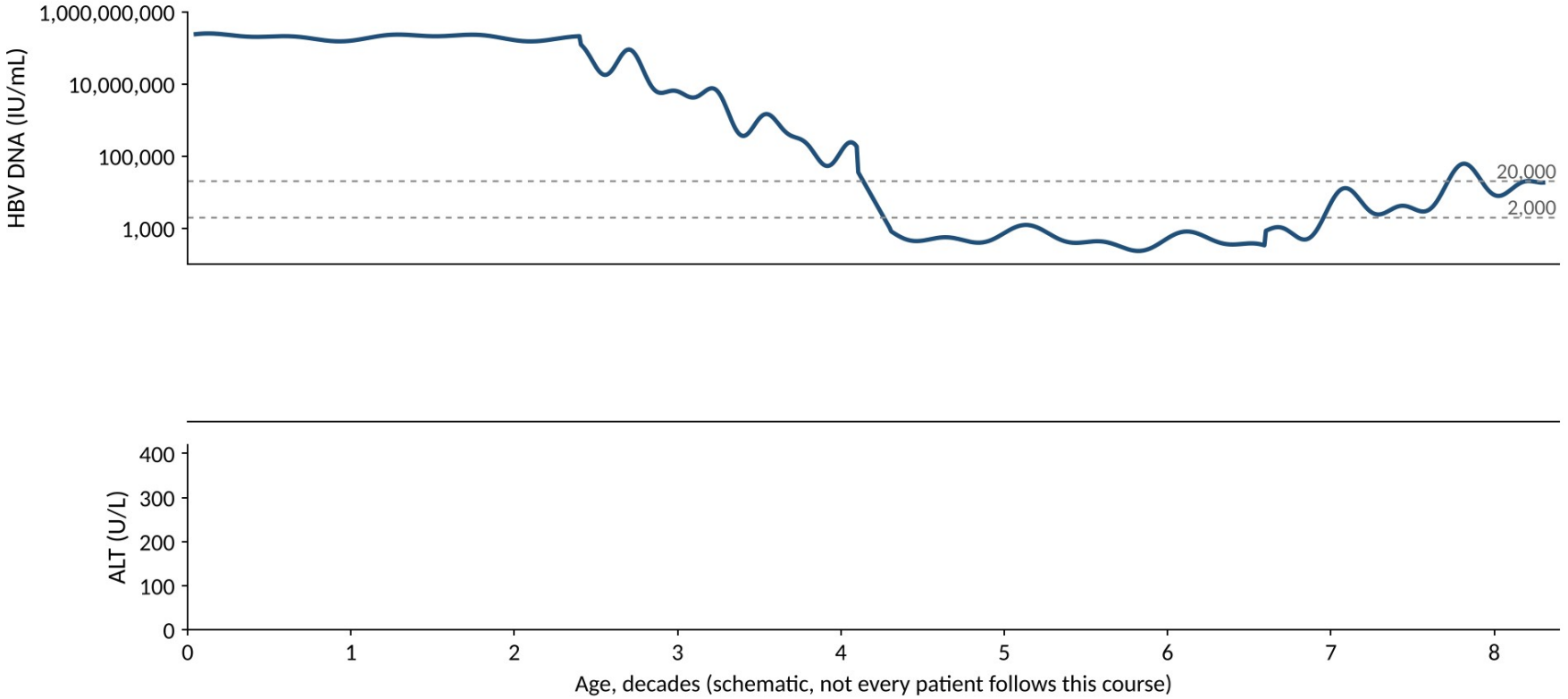
CDC 2023 (Table 1), PMID 36893044; AASLD 2018 hepatitis B guidance, PMID 29405329.

Natural history of chronic hepatitis B

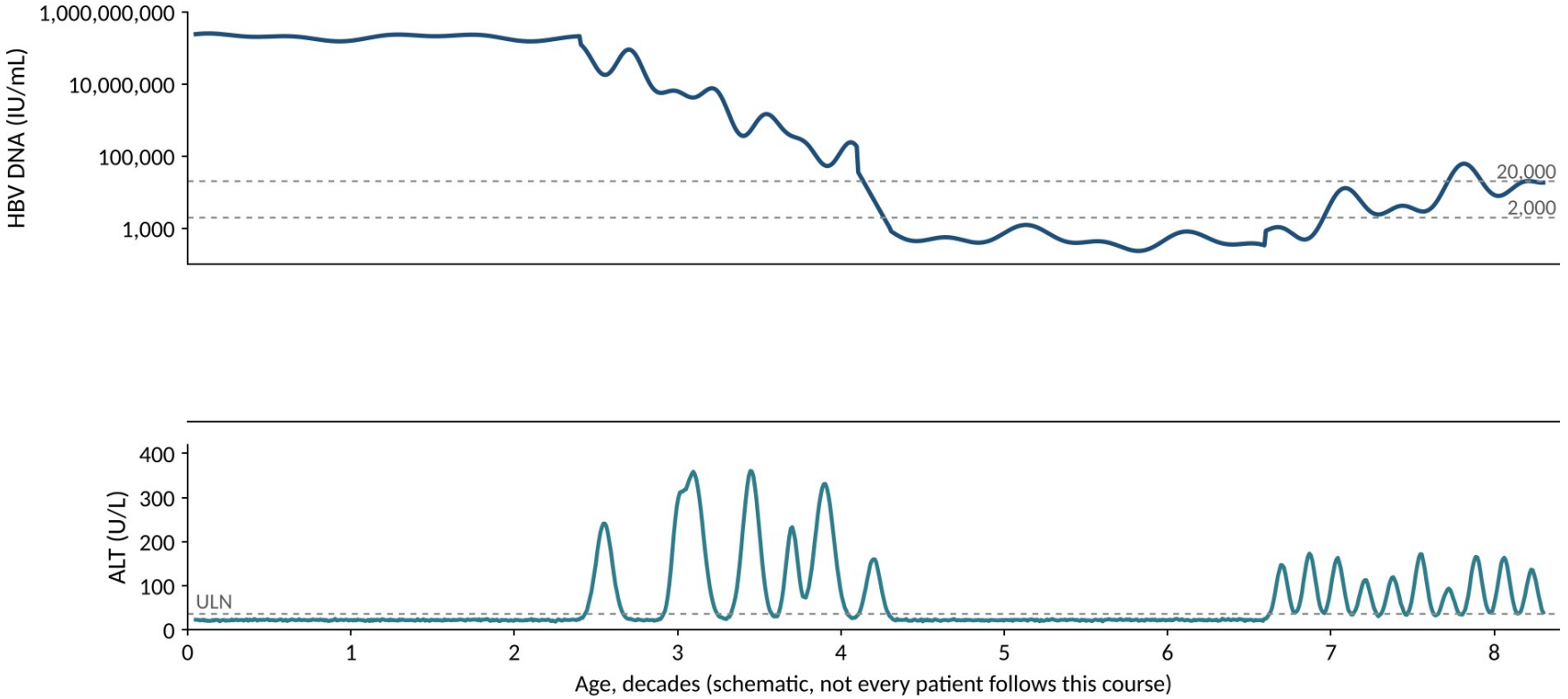
Where do HBV DNA, ALT and the antigens go over a lifetime?



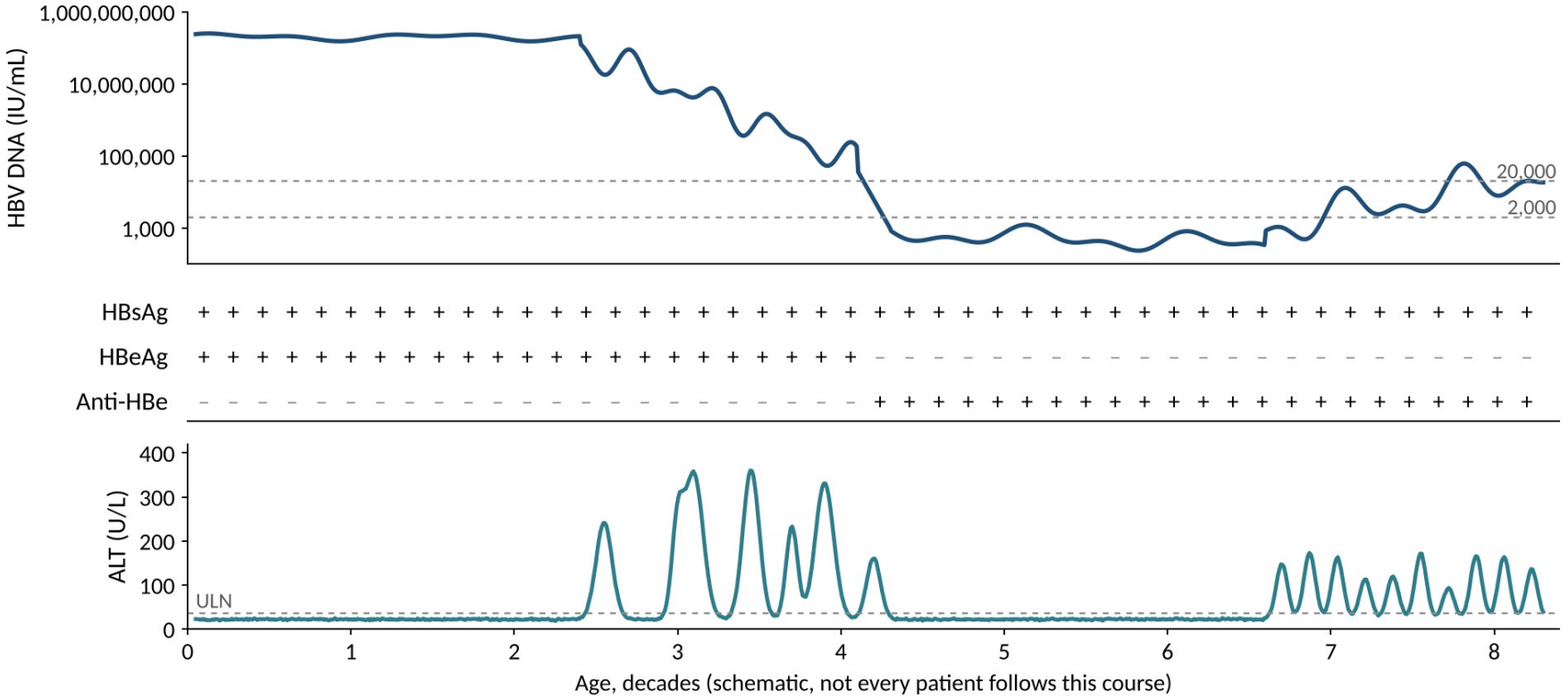
Natural history of chronic hepatitis B



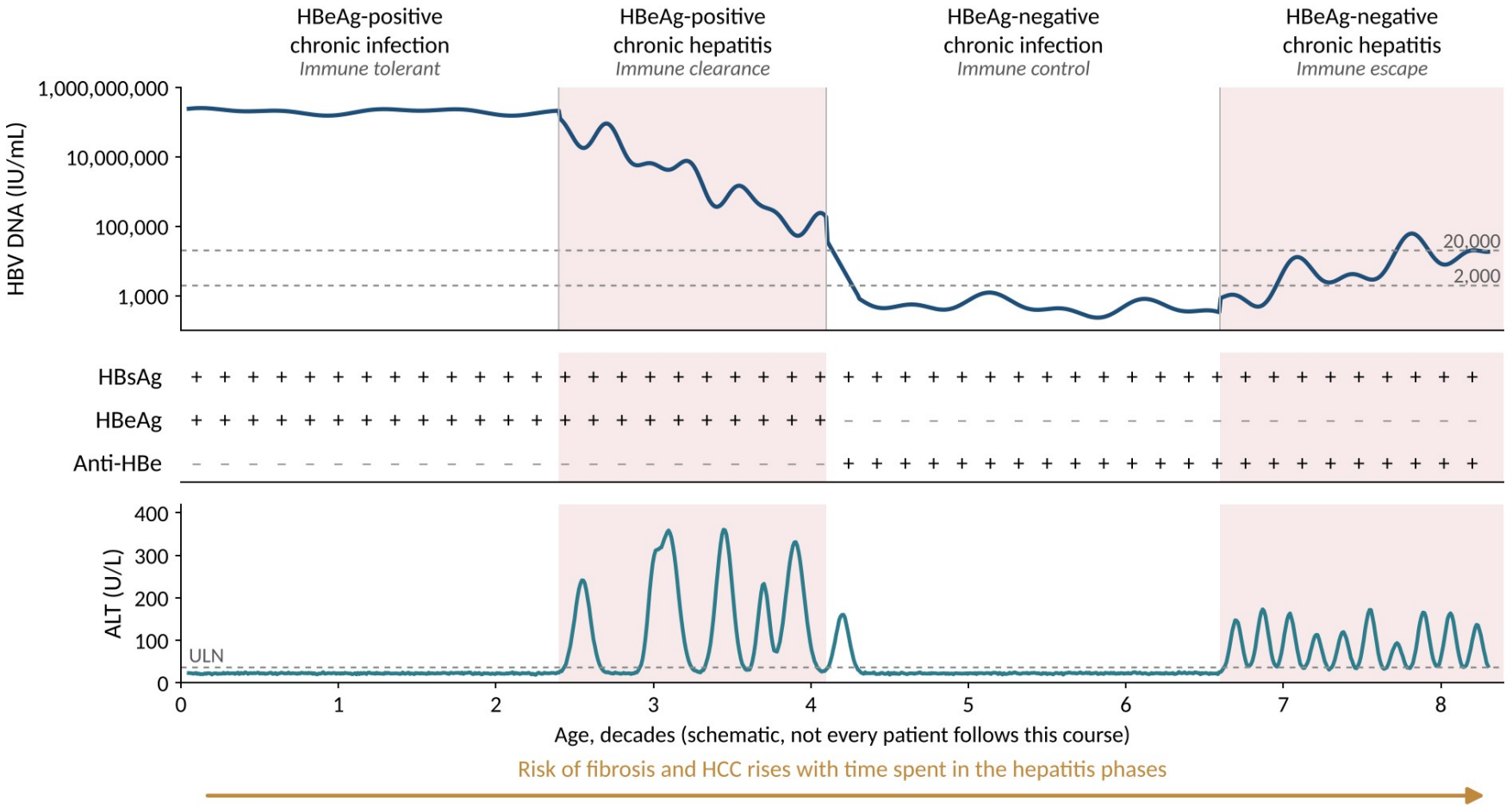
Natural history of chronic hepatitis B



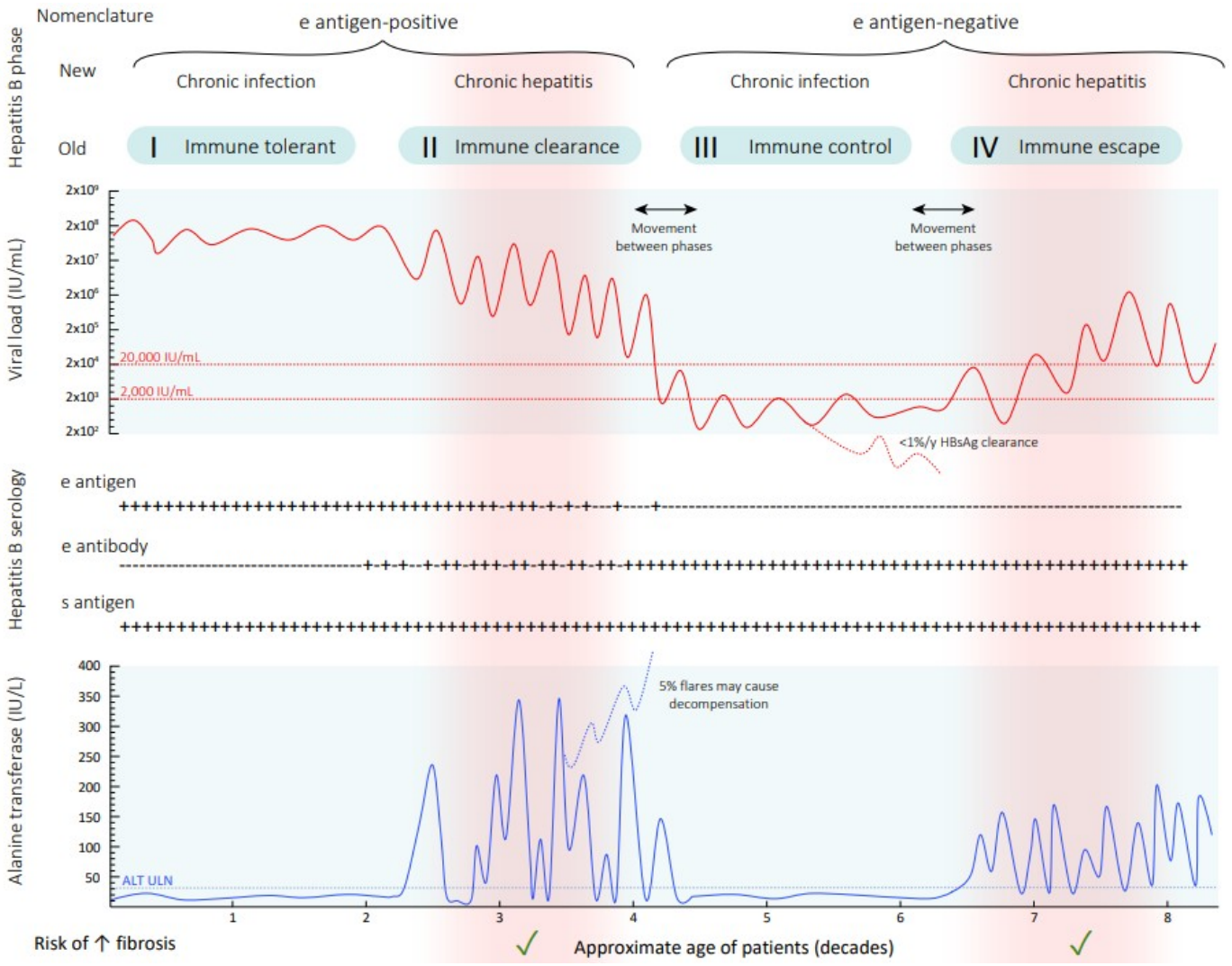
Natural history of chronic hepatitis B



Natural history of chronic hepatitis B



The published version



GESA Australian consensus recommendations for the management of hepatitis B infection, 2022, Figure 3.1 (as shown in the ASHM B Positive chapter).

Match the labs to the phase

HBeAg	HBV DNA (IU/mL)	ALT	Phase
Positive	10,000,000 or more	Normal	
Positive	10,000 to 10,000,000	Raised	
Negative	Below 2,000	Normal	
Negative	Above 2,000	Raised	

EASL Clinical Practice Guidelines on hepatitis B 2025, Table 4. J Hepatol 2025;83:502-583, PMID 40348683. Values are 'usually'.

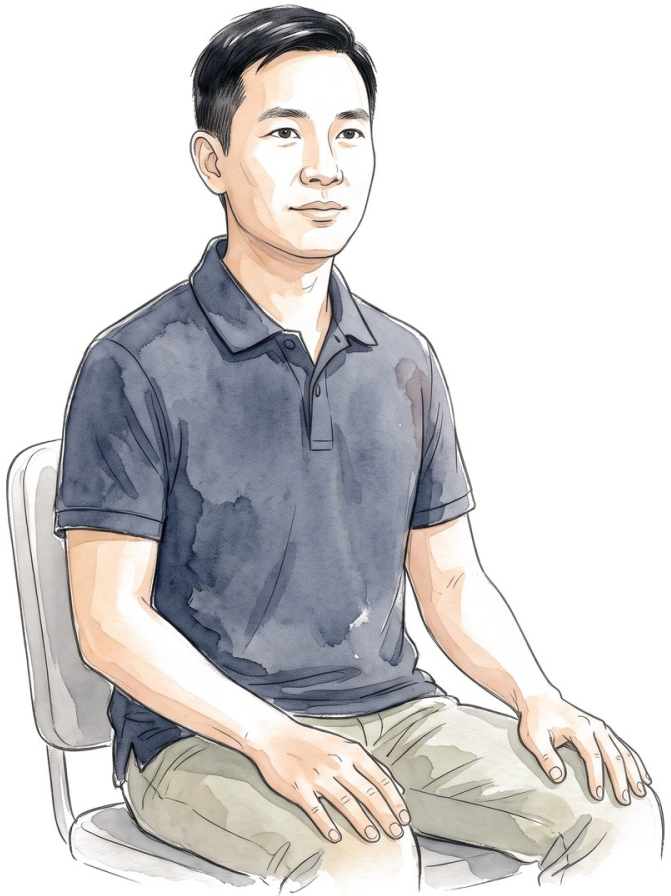
Match the labs to the phase

HBeAg	HBV DNA (IU/mL)	ALT	Phase
Positive	10,000,000 or more	Normal	HBeAg-positive chronic infection
Positive	10,000 to 10,000,000	Raised	HBeAg-positive chronic hepatitis
Negative	Below 2,000	Normal	HBeAg-negative chronic infection
Negative	Above 2,000	Raised	HBeAg-negative chronic hepatitis

EASL Clinical Practice Guidelines on hepatitis B 2025, Table 4. J Hepatol 2025;83:502-583, PMID 40348683. Values are 'usually'.

34-year-old man, born in Vietnam

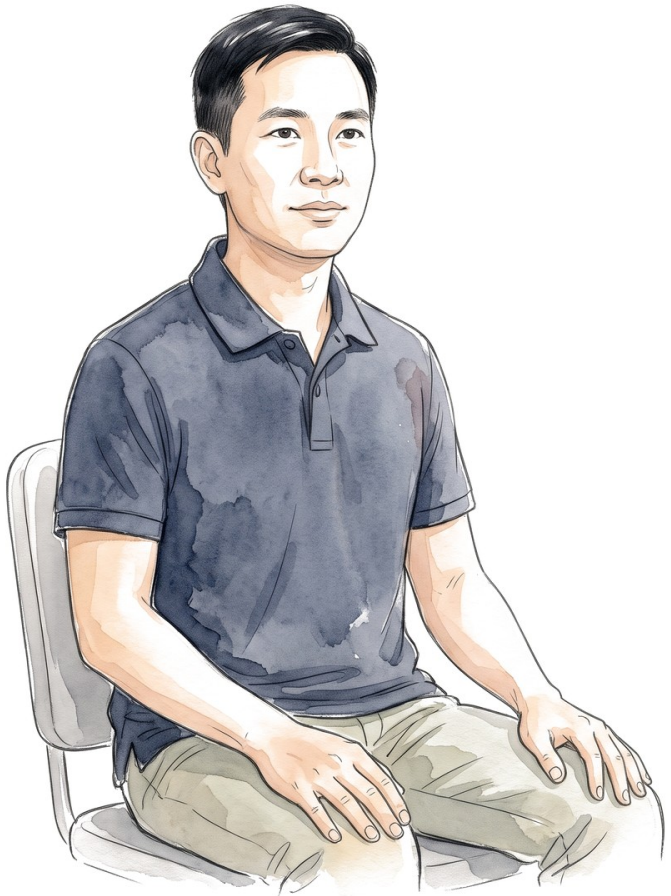
Test	Result
HBsAg	Positive
HBeAg	Positive
Anti-HBe	Negative
HBV DNA	60,000,000 IU/mL
ALT (ULN 35)	24 U/L
Platelets	245 x 10^9/L
Liver stiffness	5.1 kPa



Do you treat him?

34-year-old man, born in Vietnam

Test	Result
HBsAg	Positive
HBeAg	Positive
Anti-HBe	Negative
HBV DNA	60,000,000 IU/mL
ALT (ULN 35)	24 U/L
Platelets	245 x 10^9/L
Liver stiffness	5.1 kPa

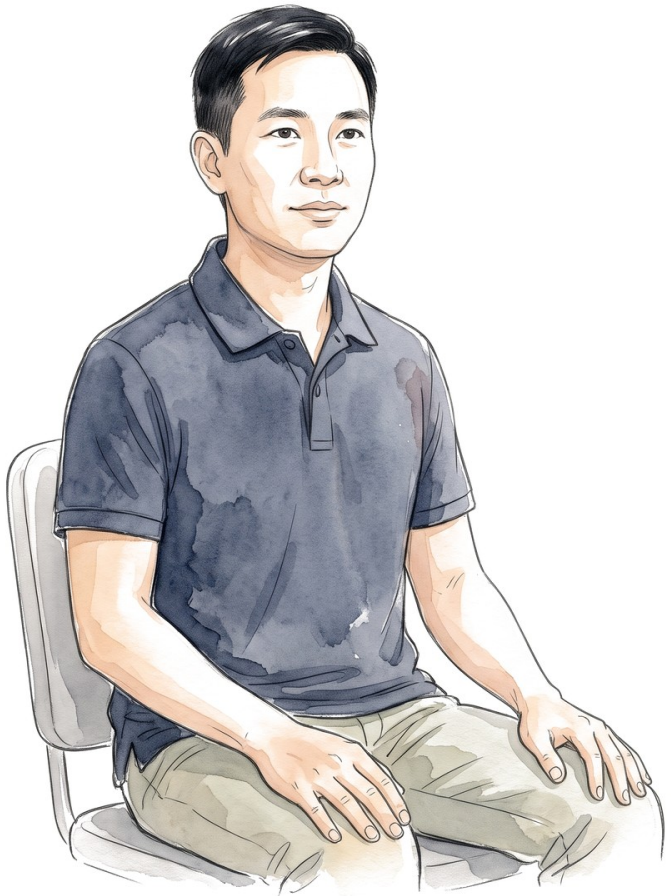


Do you treat him?

No. Monitor ALT and DNA every 3 to 6 months

Same man, now 41 years old

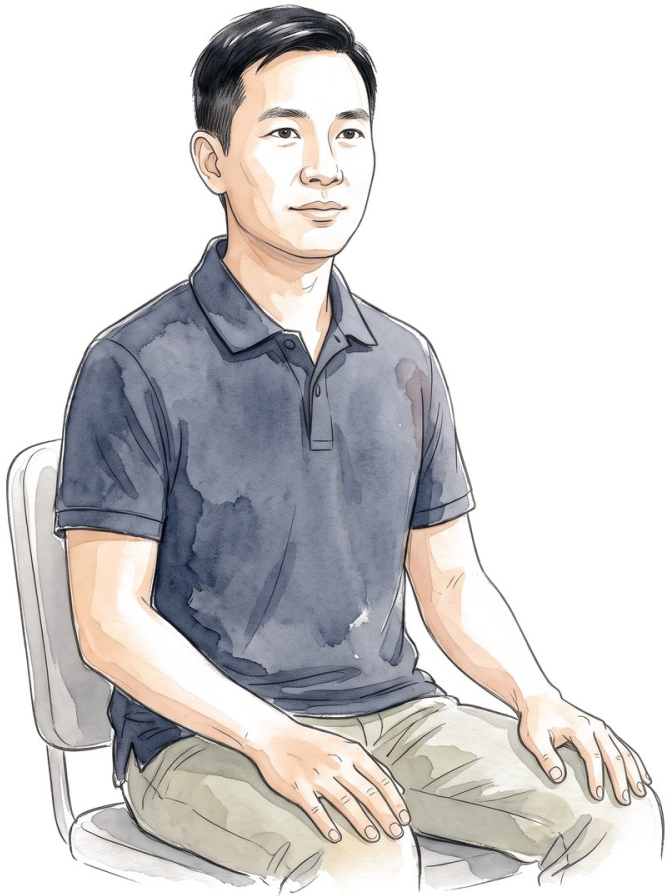
Test	Result
HBsAg	Positive
HBeAg	Positive
Anti-HBe	Negative
HBV DNA	55,000,000 IU/mL
ALT (ULN 35)	26 U/L
Platelets	245 x 10^9/L
Liver stiffness	5.1 kPa



Does the answer change?

Same man, now 41 years old

Test	Result
HBsAg	Positive
HBeAg	Positive
Anti-HBe	Negative
HBV DNA	55,000,000 IU/mL
ALT (ULN 35)	26 U/L
Platelets	245 x 10^9/L
Liver stiffness	5.1 kPa

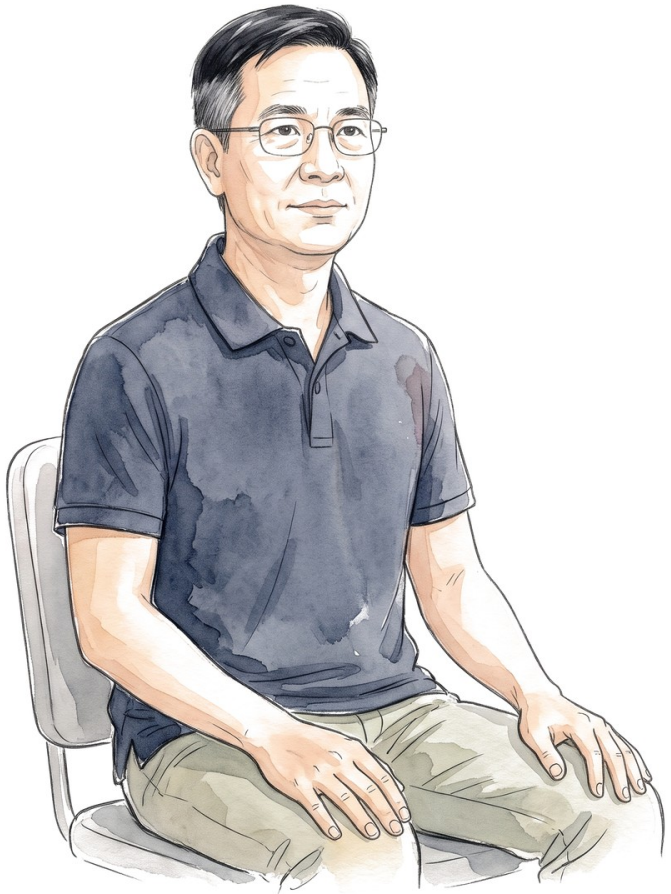


Does the answer change?

AASLD/IDSA 2025: over age 40 supports treatment

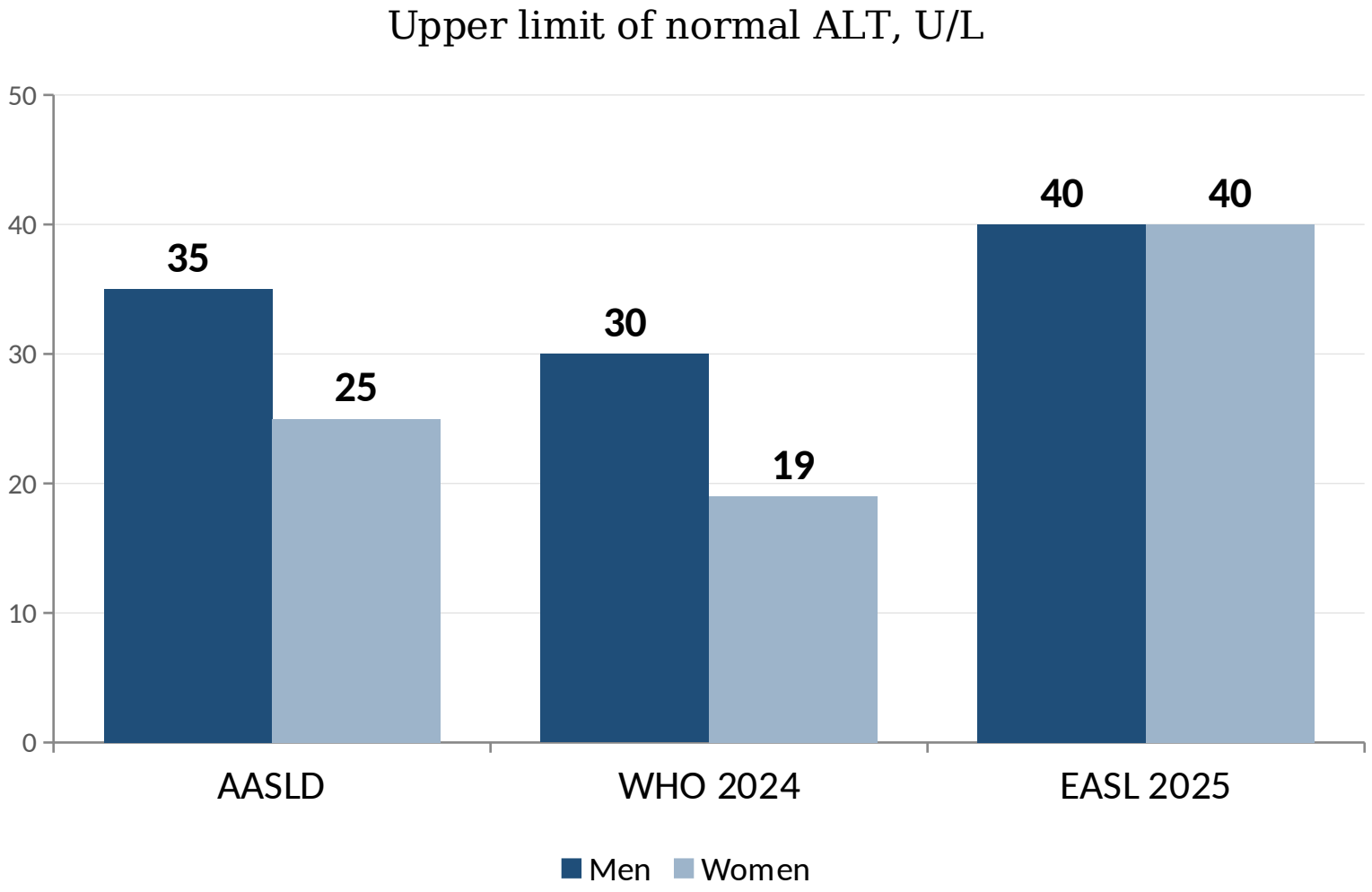
Same man, now 52 years old

Test	Result
HBsAg	Positive
HBeAg	Negative
Anti-HBe	Positive
HBV DNA	8,000 IU/mL
ALT	48 U/L
Platelets	190 x 10^9/L
FIB-4	1.8
Liver stiffness	6.8 kPa



Do you treat him?

What counts as a normal ALT?



His ALT

48 U/L

Above all three limits

Upper limits of normal as listed in the guidelines. AASLD 2018 PMID 29405329; WHO 2024; EASL 2025 PMID 40348683.

Same labs, four guidelines

Treat or not?

	AASLD 2018	AASLD/IDSA 2025	EASL 2025	WHO 2024
52 y, HBeAg-negative, DNA 8,000, ALT 48				
Compensated cirrhosis, DNA 500				
Decompensated cirrhosis, any DNA				

AASLD 2018 PMID 29405329; AASLD/IDSA 2025 Hepatology 2026;83:974-997 PMID 41186418; EASL 2025 PMID 40348683; WHO 2024 (NBK614987).

Same labs, four guidelines

Treat or not?

	AASLD 2018	AASLD/IDSA 2025	EASL 2025	WHO 2024
52 y, HBeAg-negative, DNA 8,000, ALT 48	Observe	Shared decision	**Treat**	**Treat**
Compensated cirrhosis, DNA 500	Treat	Treat	Treat	Treat
Decompensated cirrhosis, any DNA	Treat	Treat	Treat	Treat

AASLD 2018 PMID 29405329; AASLD/IDSA 2025 Hepatology 2026;83:974-997 PMID 41186418; EASL 2025 PMID 40348683; WHO 2024 (NBK614987).

Does treating this group help?

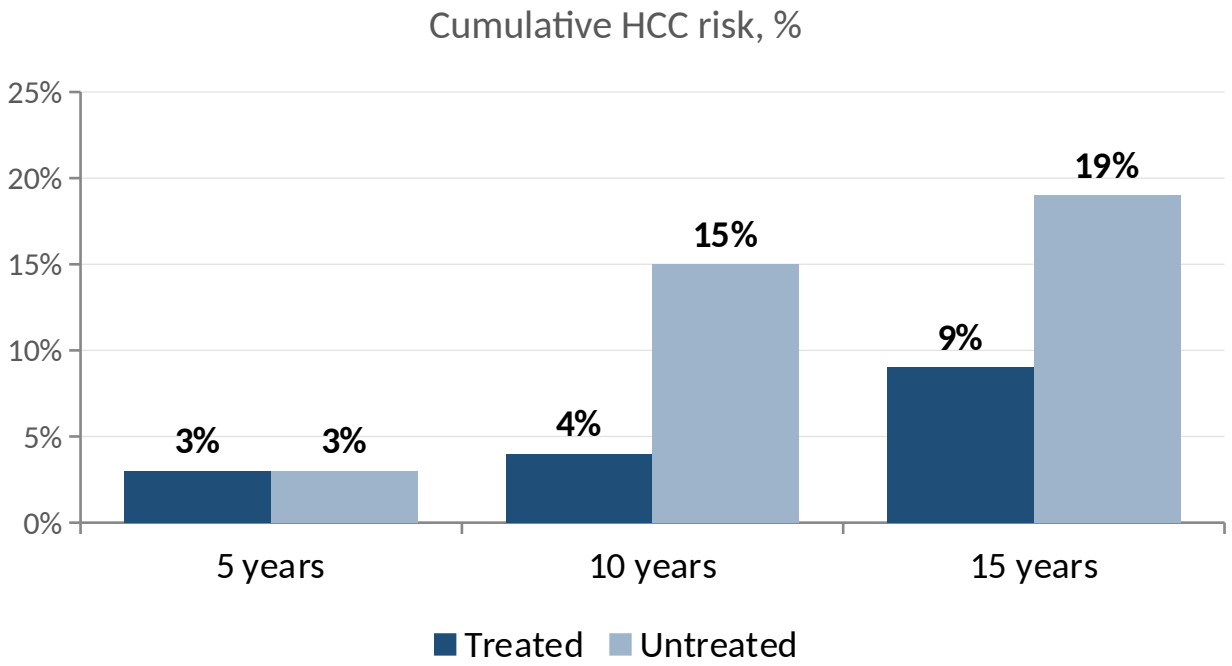
Study	Design	Patients	Outcome
Huang 2023	Observational	855, no advanced fibrosis	HCC, 5 to 15 years
TORCH-B 2021	Randomized	160, ALT 1 to 2 x ULN	Progression at 3 years

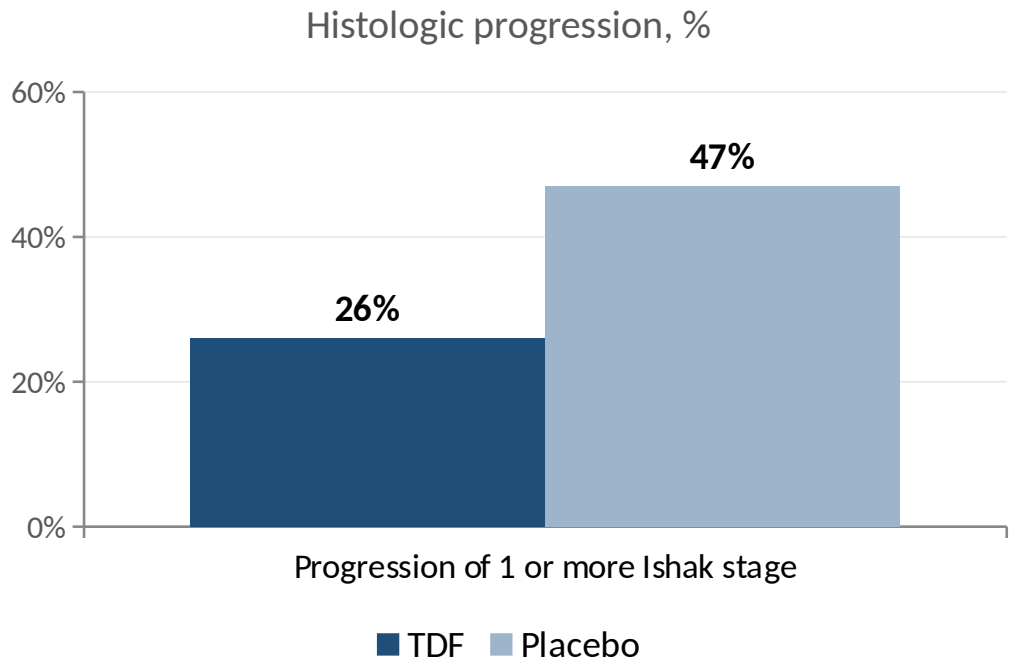
What do you expect?

Huang DQ et al. Hepatology 2023;78:1558-1568, PMID 37184202. Hsu YC et al. Lancet Infect Dis 2021;21:823-833, PMID 33524314.

Does treating this group help?

Study	Design	Patients	Outcome
Huang 2023	Observational	855, no advanced fibrosis	HCC, 5 to 15 years
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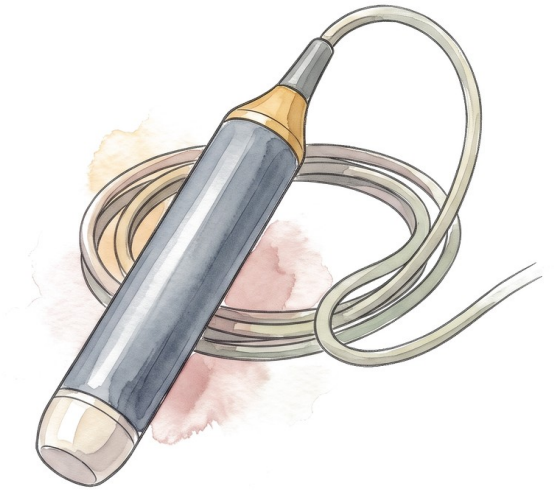




Huang DQ et al. Hepatology 2023;78:1558-1568, PMID 37184202. Hsu YC et al. Lancet Infect Dis 2021;21:823-833, PMID 33524314.

Fibrosis staging by elastography

	EASL 2025	WHO 2024
Significant fibrosis (F2)	Above 7 kPa	Above 7 kPa, or APRI above 0.5
Advanced fibrosis (F3)	Above 8 kPa	
Cirrhosis (F4)	Above 11 kPa	APRI above 2 in the 2015 guideline



EASL 2025 PMID 40348683; WHO 2024 chapter 5.

Treat everyone with high DNA and normal ALT?

	Kim, Gut 2018	Jeon, Clin Mol Hepatol 2021
Patients	413 untreated immune-tolerant, 1,497 treated immune-active	Biomarker-defined immune tolerance, FIB-4 under 1.45 subgroup
Result	10-year HCC 12.7% vs 6.1%; adjusted HR 2.54 (1.54 to 4.18)	HCC 0% at 1, 3 and 5 years vs 0.2, 0.6 and 1.4% in comparators
Limit	Different phases, selected treatment	Small low-risk sample

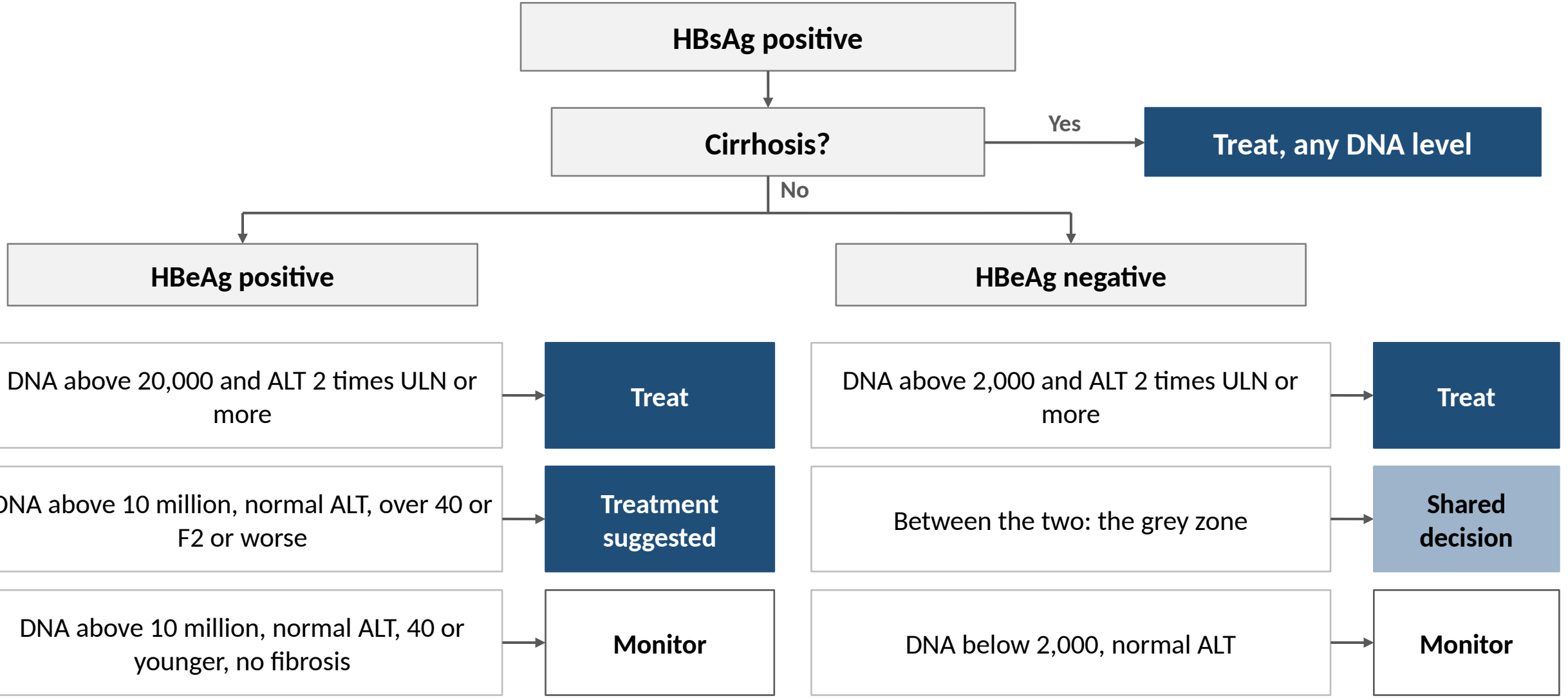
Kim GA et al. Gut 2018;67:945-952, PMID 29055908. Jeon MY et al. Clin Mol Hepatol 2021;27:295-304, PMID 33317247.

The indeterminate phase

	Result
Meta-analysis, 103 studies (Lai, J Hepatol 2025)	Significant fibrosis in 39.7%; HCC 0.32% a year; treatment IRR 0.38 (0.18 to 0.79)
ATTENTION interim, 734 patients (Lim, 2025)	Events 2 vs 9, HR 0.21, 97.5% CI 0.04 to 1.20; stopping boundary not crossed

Lai JC et al. J Hepatol 2025;82:992-1003, PMID 39577468. Lim YS et al. Lancet Gastroenterol Hepatol 2025;10:295-305, PMID 39914435.

Who to treat: summary



AASLD 2018 (PMID 29405329); AASLD/IDSA 2025 R3, R4 and Figure 5 (PMID 41186418). ALT ULN 35 men, 25 women.

58-year-old man, eGFR 45

Which nucleos(t)ide analogue?

	Dose	At eGFR 30 to 49	
Entecavir	0.5 mg daily	0.5 mg every 48 hours	
TDF	300 mg daily	300 mg every 48 hours	
TAF	25 mg daily with food	25 mg daily	

US prescribing information: Baraclude, Viread, Vemlidy (DailyMed). Renal doses use creatinine clearance.

58-year-old man, eGFR 45

Which nucleos(t)ide analogue?

	Dose	At eGFR 30 to 49	
Entecavir	0.5 mg daily	0.5 mg every 48 hours	Dose-adjust
TDF	300 mg daily	300 mg every 48 hours	Dose-adjust, kidney and bone effects
TAF	25 mg daily with food	25 mg daily	**No adjustment**

TAF, or entecavir with the dose adjusted

US prescribing information: Baraclude, Viread, Vemlidy (DailyMed). Renal doses use creatinine clearance.

TAF versus TDF at week 48

HBV DNA below 29 IU/mL	TAF	TDF
HBeAg-positive (Chan)	64%	67%
HBeAg-negative (Buti)	94%	93%

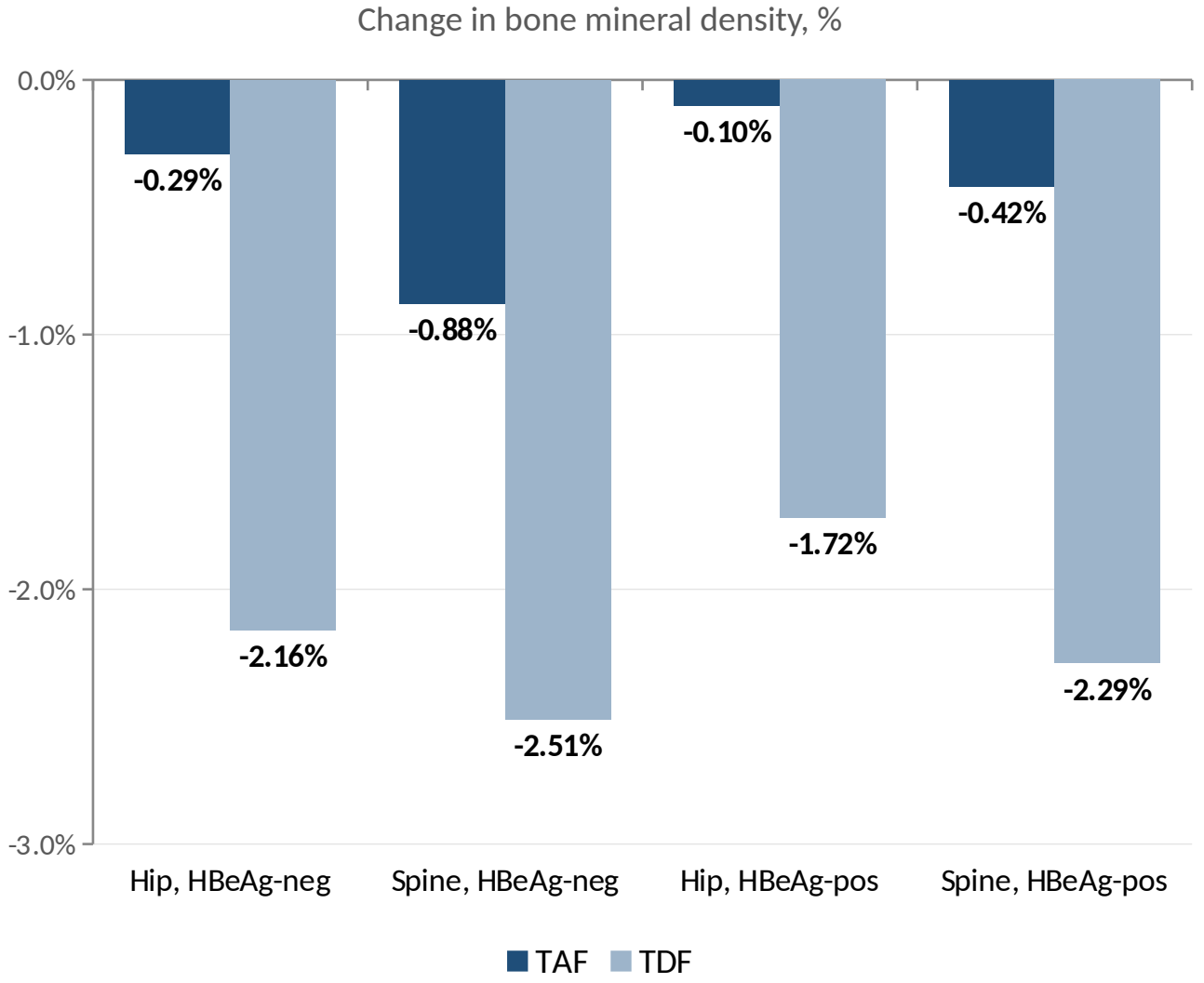
What differs?

Chan HLY et al. Lancet Gastroenterol Hepatol 2016;1:185-195 (PMID 28404091). Buti M et al. 2016;1:196-206 (PMID 28404092).

TAF versus TDF at week 48

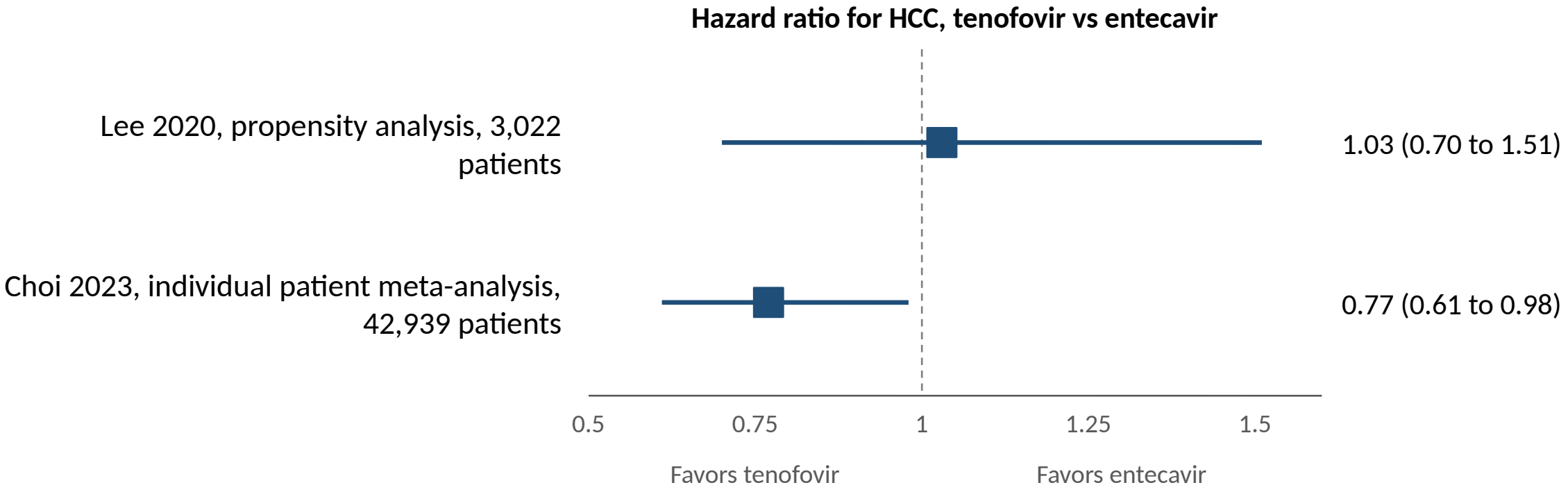
HBV DNA below 29 IU/mL	TAF	TDF
HBeAg-positive (Chan)	64%	67%
HBeAg-negative (Buti)	94%	93%

Median eGFR change in the HBeAg-negative trial: -1.8 (TAF) vs -4.8 mL/min (TDF)



Chan HLY et al. Lancet Gastroenterol Hepatol 2016;1:185-195 (PMID 28404091). Buti M et al. 2016;1:196-206 (PMID 28404092).

Entecavir or tenofovir for HCC?



Yoo 2025, Korea, TAF vs TDF: 12.38 vs 15.39 HCC per 1,000 person-years, 17,537 in each matched group

Lee SW 2020 (PMID in dossier ref 31); Choi J et al. 2023 (dossier ref 32); Yoo 2025 (dossier ref 28).

Pegylated interferon

Strategy	Trial	Result
Switch from entecavir, HBeAg-positive	OSST, 200 patients	HBsAg loss 8.5% vs 0% at week 48
Add to TDF for 48 weeks	Marcellin, 740 patients	HBsAg loss 9.1% (48 weeks), 2.8% (16 weeks), 0% (TDF alone), 2.8% (PegIFN alone) at week 72

Ning Q et al. 2014 (dossier ref 34); Marcellin P et al. 2016 (dossier ref 35).

Monitoring

Setting	Schedule
On a nucleos(t)ide analogue	DNA every 3 months until undetectable, then every 3 to 6 months
On TDF	Creatinine, phosphate, urine glucose and protein at baseline and at least yearly
Untreated, HBeAg-negative infection	ALT every 3 months in year 1, then every 6 to 12 months; HBsAg yearly
HCC surveillance	Ultrasound plus AFP every 6 months when eligible, also with suppressed DNA

AASLD 2018, PMID 29405329; EASL 2025, PMID 40348683.

Choosing the drug: summary

Situation	Drug
Most patients	Entecavir, TDF or TAF
Kidney or bone disease, or at risk	TAF, or entecavir with the dose adjusted
Prior lamivudine resistance	TDF or TAF, not entecavir
Pregnancy	TDF (most data) or TAF
HIV	TDF or TAF with emtricitabine or lamivudine, inside the ART regimen
Decompensated cirrhosis	Entecavir or TDF. The US TAF label does not recommend Child-Pugh B or C
Wants a finite course, compensated	Pegylated interferon, in selected patients

Calculating PAGE-B

Variable	Category	Points
Age, years	16 to 29	0
	30 to 39	2
	40 to 49	4
	50 to 59	6
	60 to 69	8
	70 or older	10
Sex	Female	0
	Male	6
Platelets, x 10^9/L	200 or more	0
	100 to 199	6
	Under 100	9

52-year-old man

Started on TAF

Platelets 190

What is his PAGE-B score?

Points as printed in EASL 2025 Table 13. PAGE-B: Papatheodoridis GV et al. J Hepatol 2016.

Calculating PAGE-B

Variable	Category	Points
Age, years	16 to 29	0
	30 to 39	2
	40 to 49	4
	50 to 59	6
	60 to 69	8
	70 or older	10
Sex	Female	0
	Male	6
Platelets, x 10^9/L	200 or more	0
	100 to 199	6
	Under 100	9

52-year-old man

Started on TAF

Platelets 190

6 + 6 + 6 = 18

High risk: 18 or more

5-year HCC about 16 to 17%

Points as printed in EASL 2025 Table 13. PAGE-B: Papatheodoridis GV et al. J Hepatol 2016.

Calculate it yourself

PAGE-B	5-year HCC, derivation	5-year HCC, validation
9 or less	0%	0%
10 to 17	3%	4%
18 or more	17%	16%

- PAGE-B was built in treated patients; REACH-B was built in untreated patients
- The HBsAg-based model on the site is a 2026 research model, not validated

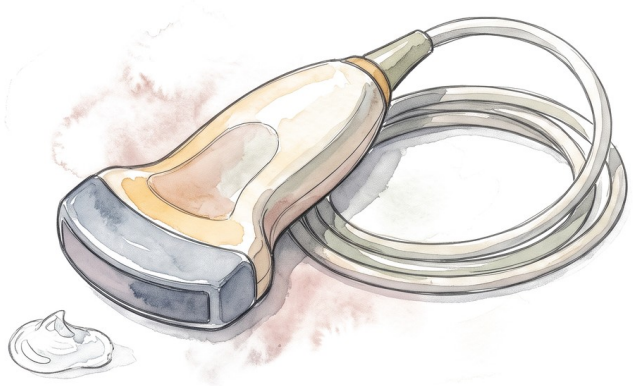
qaddodi.github.io/chb-hcc-risk

PAGE-B: Papatheodoridis GV et al. J Hepatol 2016. VA validation: Kim 2022.

Who gets surveillance?

Ultrasound and AFP every 6 months?

Patient	Surveillance
Compensated cirrhosis, Child-Pugh A	
Man from an endemic region, age 38, no cirrhosis	
Man from an endemic region, age 46, no cirrhosis	
Woman from an endemic region, age 46, no cirrhosis	
Family history of HCC	
PAGE-B score of 12	

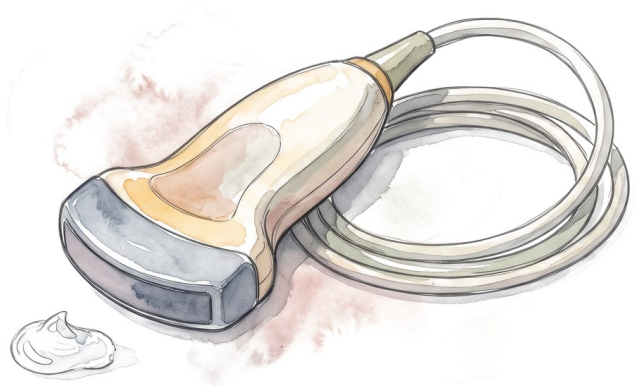


AASLD HCC practice guidance 2023, Table 1. Hepatology 2023;78:1922-1965, PMID 37199193.

Who gets surveillance?

Ultrasound and AFP every 6 months?

Patient	Surveillance
Compensated cirrhosis, Child-Pugh A	Yes
Man from an endemic region, age 38, no cirrhosis	No
Man from an endemic region, age 46, no cirrhosis	Yes
Woman from an endemic region, age 46, no cirrhosis	No
Family history of HCC	Yes
PAGE-B score of 12	Yes



AASLD HCC practice guidance 2023, Table 1. Hepatology 2023;78:1922-1965, PMID 37199193.

Which HCC score for which patient

Score	Built in	Variables
PAGE-B	1,815 treated White Europeans	Age, sex, platelets
mPAGE-B	3,001 treated Asian patients	Age, sex, platelets, albumin
REACH-B	3,584 untreated Taiwanese, noncirrhotic	Sex, age, ALT, HBeAg, DNA
aMAP	17,374, mostly treated Asian HBV	Age, sex, albumin-bilirubin, platelets
REAL-B	8,048 treated Asian patients	Sex, age, alcohol, diabetes, cirrhosis, platelets, AFP

Papatheodoridis 2016; Kim 2018; Yang 2011; Fan 2020; Yang 2020; Kim 2022 (dossier refs 38-42, 46).

Does qHBsAg refine HCC risk?

Tseng, Gastroenterology 2012	Result
2,688 noncirrhotic Taiwanese, 191 HCC, mean 14.7 years	HBeAg-negative with DNA under 2,000: HBsAg 1,000 or more vs under 1,000 gave adjusted HR 13.7 (4.8 to 39.3)
Whole cohort	DNA was the stronger predictor; HBsAg alone did not add independently

Tseng TC et al. Gastroenterology 2012 (dossier ref 48).

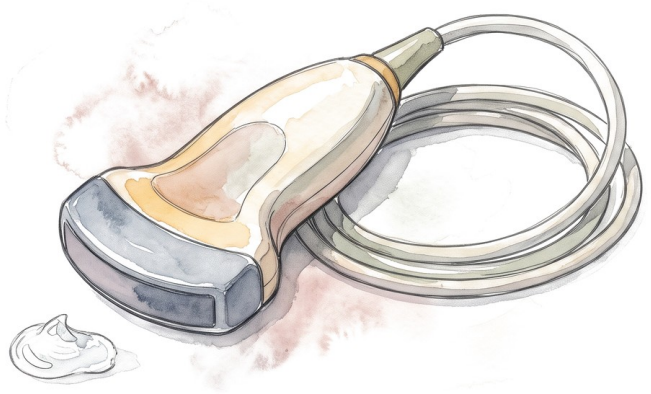
Integration of HBV DNA

Study	Finding
Mason 2016, 26 patients aged 14 to 39	Integration and clonal hepatocyte expansion found in immune-tolerant patients as well, before ALT rises
Grudda 2022, 16 HIV/HBV patients	Integrated DNA keeps producing HBsAg despite antiviral therapy

Mason WS et al. Gastroenterology 2016 (dossier ref 50); Grudda T et al. 2022 (ref 51).

HCC surveillance: summary

- **Ultrasound and AFP every 6 months**
- Cirrhosis
- No cirrhosis, from an endemic region: men over 40, women over 50; people of African origin earlier
- Family history of HCC
- PAGE-B 10 or more
- Hepatitis D coinfection
- **Keep going after HBsAg loss if:**
- Cirrhosis or a family history of HCC, or HBsAg lost after 40 in men and after 50 in women



AASLD HCC guidance 2023, Table 1 (PMID 37199193); AASLD/IDSA 2025 R6 to R9 (PMID 41186418).

Risk of HBV reactivation

How risky is each drug if HBsAg is negative and anti-HBc is positive?

Drug	Reactivation risk
Rituximab or other anti-CD20	
Anthracyclines	
JAK inhibitors	
Prednisone 20 mg or more for 4 weeks or longer	
Anti-TNF	
Immune checkpoint inhibitors	
Methotrexate, azathioprine	

AGA guideline 2025, Table 6. Gastroenterology 2025;168:267-284, PMID 39863345. High above 10%, moderate 1 to 10%, low below 1%.

Risk of HBV reactivation

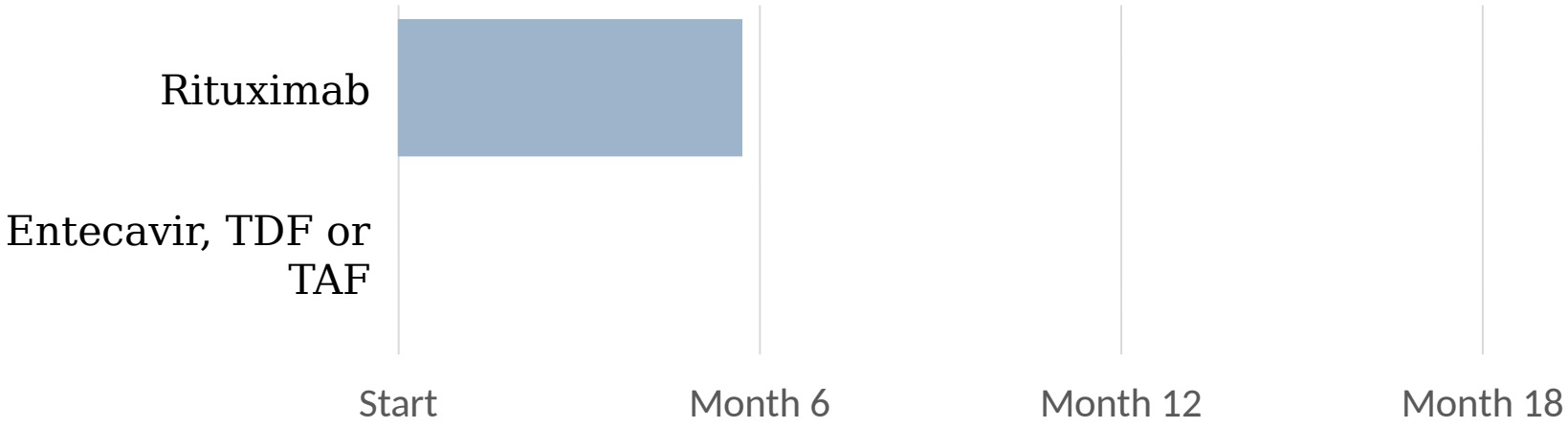
How risky is each drug if HBsAg is negative and anti-HBc is positive?

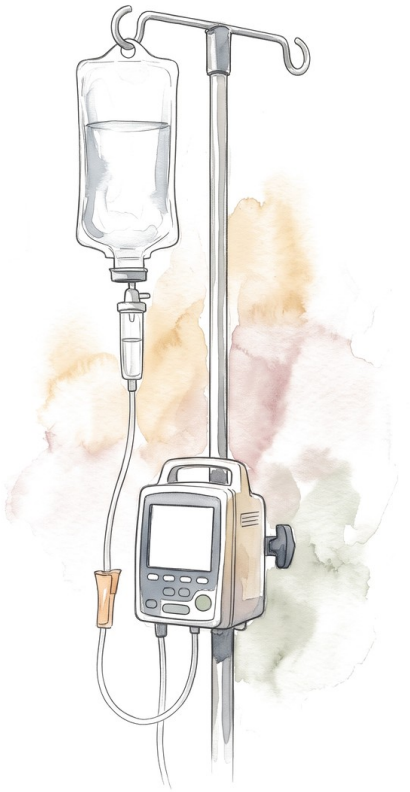
Drug	Reactivation risk
Rituximab or other anti-CD20	**High**
Anthracyclines	Moderate
JAK inhibitors	Moderate
Prednisone 20 mg or more for 4 weeks or longer	Moderate
Anti-TNF	Low
Immune checkpoint inhibitors	Low
Methotrexate, azathioprine	Low

AGA guideline 2025, Table 6. Gastroenterology 2025;168:267-284, PMID 39863345. High above 10%, moderate 1 to 10%, low below 1%.

Rituximab, six cycles

What do you give, and for how long?

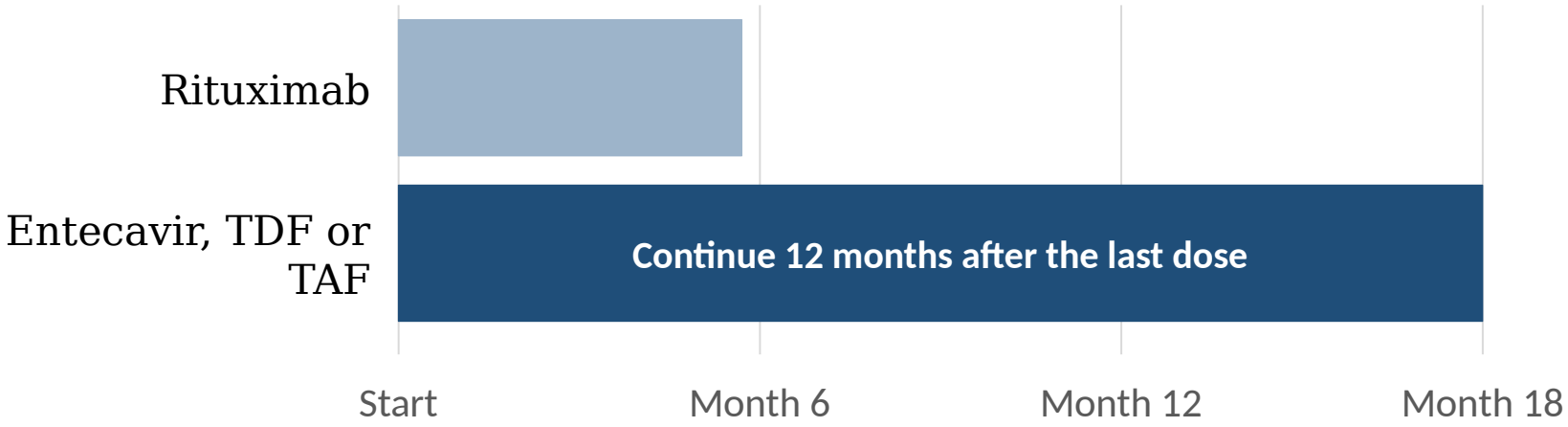


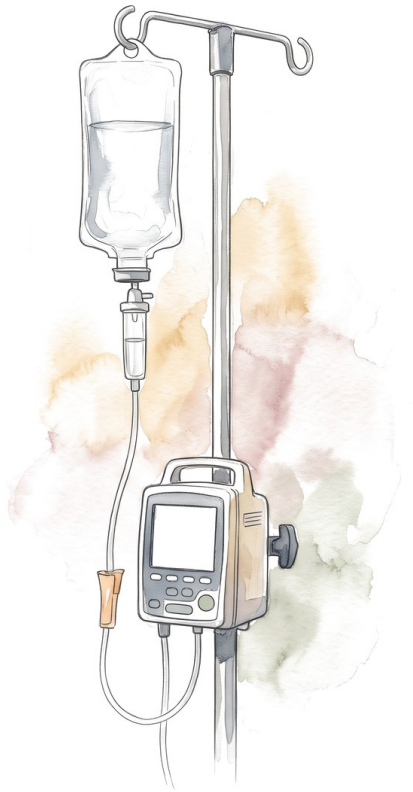


AGA 2025 (PMID 39863345); AASLD 2018 (PMID 29405329); EASL 2025 (PMID 40348683).

Rituximab, six cycles

What do you give, and for how long?

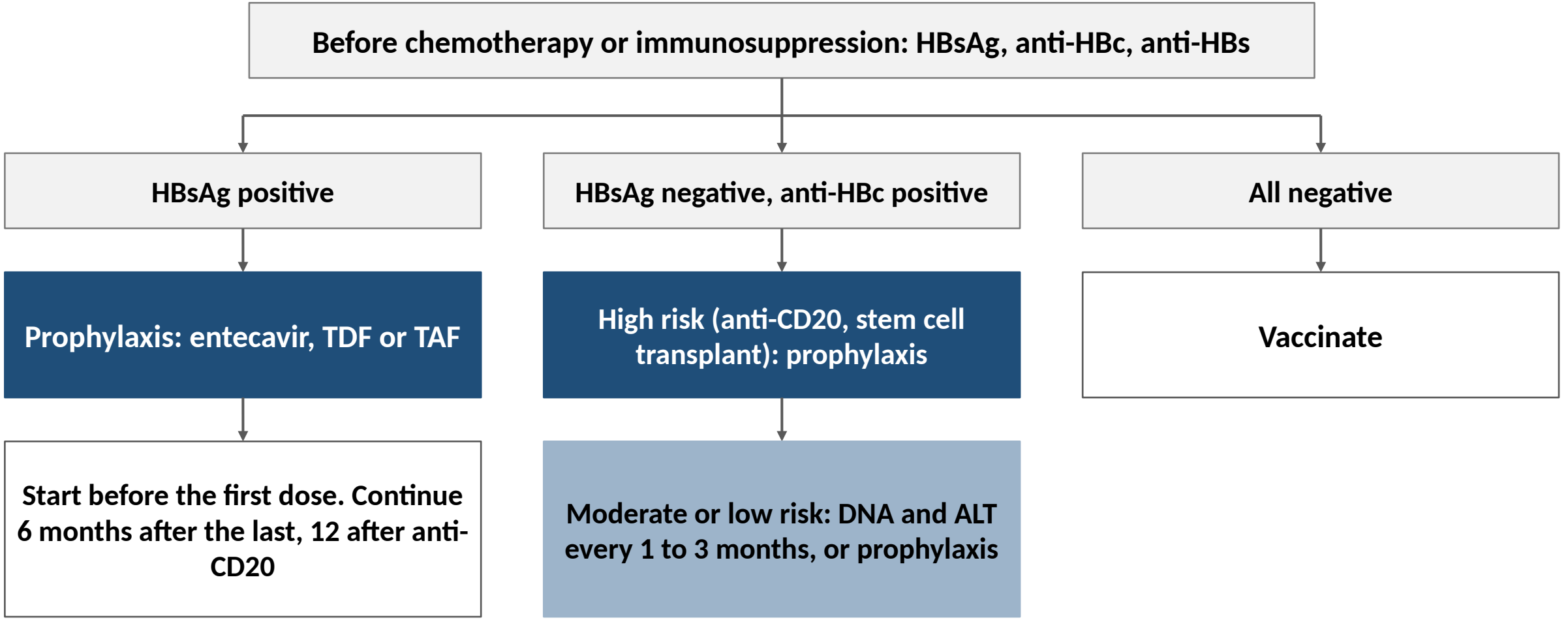




Start before the first dose. If you only monitor: DNA and ALT every 1 to 3 months

AGA 2025 (PMID 39863345); AASLD 2018 (PMID 29405329); EASL 2025 (PMID 40348683).

Before immunosuppression: summary



AGA 2025 (PMID 39863345); AASLD 2018 (PMID 29405329); EASL 2025 (PMID 40348683).

29-year-old woman, 20 weeks pregnant

HBsAg	HBeAg	HBV DNA	ALT
Positive	Positive	5,000,000 IU/mL	22 U/L

What do you do for her and for the baby?



29-year-old woman, 20 weeks pregnant

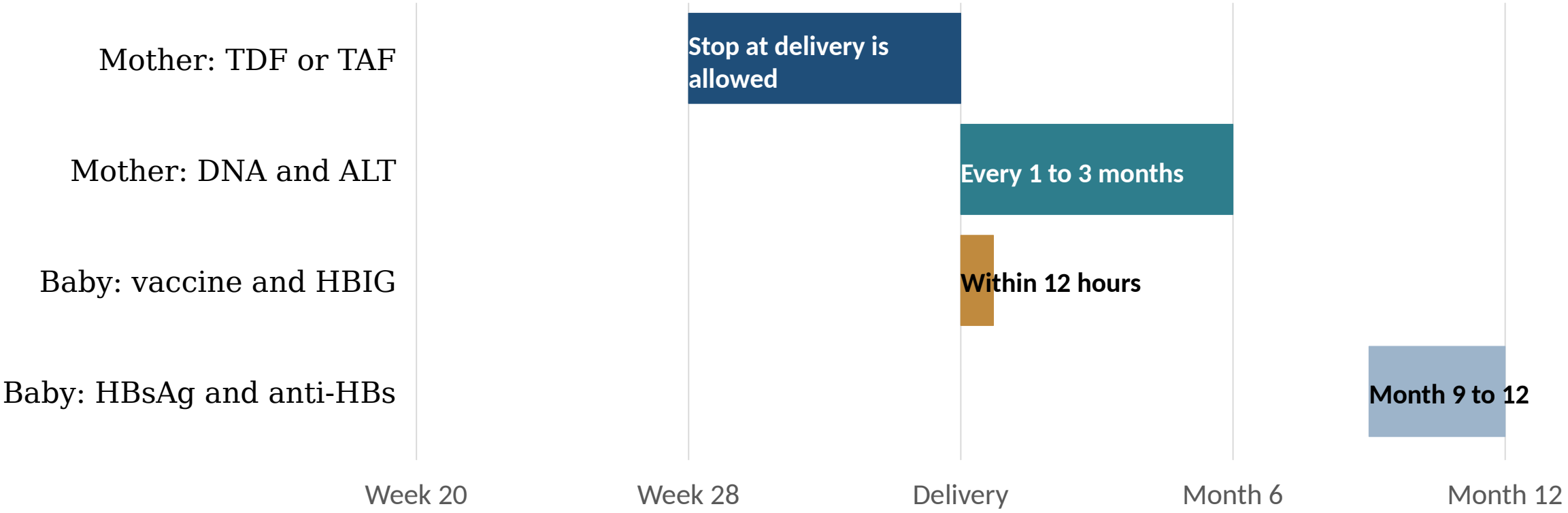
HBsAg	HBeAg	HBV DNA	ALT
Positive	Positive	5,000,000 IU/mL	22 U/L

What do you do for her and for the baby?

- TDF or TAF from week 28, because DNA is above 200,000 IU/mL
- Baby: hepatitis B vaccine and HBIG within 12 hours of birth

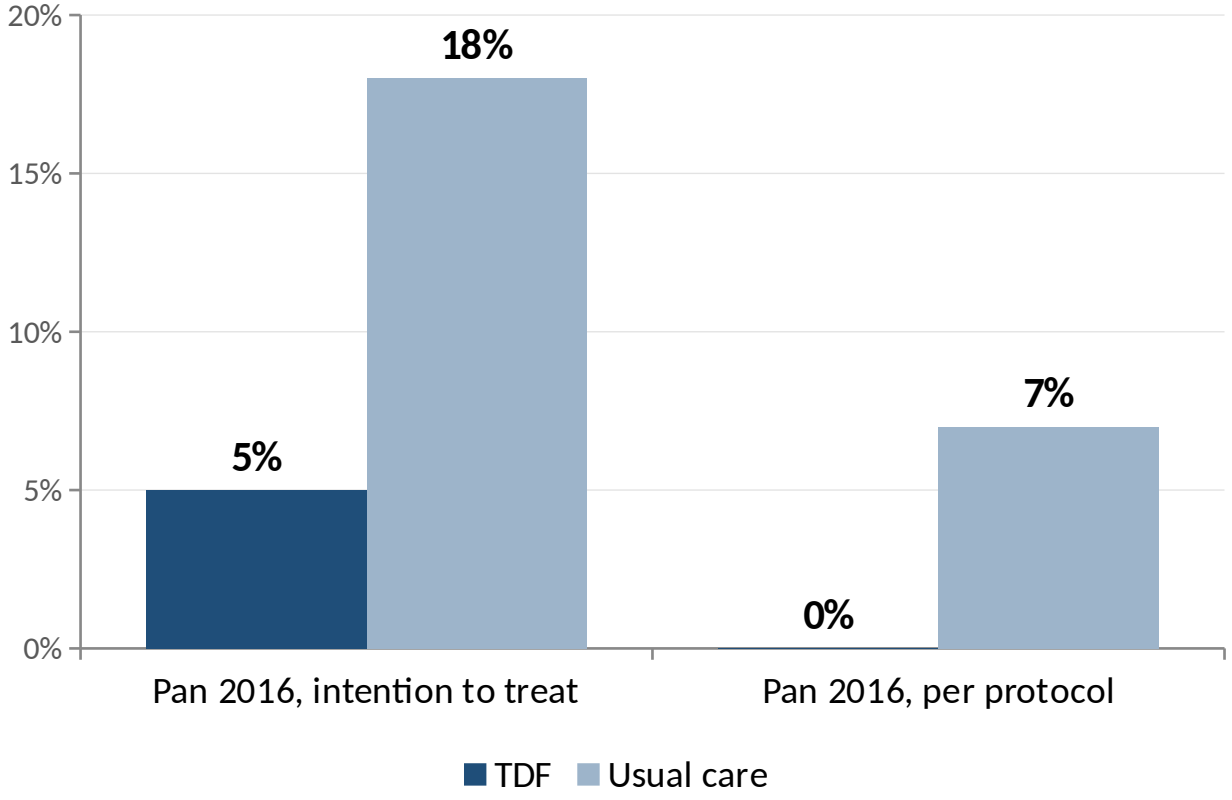


Her pregnancy and the baby's first year



AASLD/IDSA 2025 R1 (PMID 41186418); CDC perinatal hepatitis B overview, 2025.

Tenofovir in pregnancy: the trials



Jourdain 2018	TDF	Placebo
Infected infants	0 of 147	3 of 147
p	0.12	

Infant HBIG and vaccine were given about an hour after birth in Thailand

Pan CQ et al. NEJM 2016 (PMID in dossier ref 52). Jourdain G et al. NEJM 2018 (ref 53).

Same woman, after delivery

She is on TDF. Can she breastfeed?

A. No, tenofovir in breast milk harms the baby

B. Yes

C. Only after the baby finishes the vaccine series

D. Only if her DNA is undetectable



AASLD 2018 (PMID 29405329); AASLD/IDSA 2025 (PMID 41186418).

Same woman, after delivery

She is on TDF. Can she breastfeed?

A. No, tenofovir in breast milk harms the baby

B. Yes

C. Only after the baby finishes the vaccine series

D. Only if her DNA is undetectable



AASLD 2018 (PMID 29405329); AASLD/IDSA 2025 (PMID 41186418).

Pregnancy: summary

- Test HBsAg in every pregnancy
- HBsAg positive: check DNA and ALT
- **DNA above 200,000: TDF or TAF from week 28**
- Baby: hepatitis B vaccine and HBIG within 12 hours of birth, then finish the series
- Test the baby at 9 to 12 months: HBsAg and anti-HBs
- Breastfeeding is fine
- If she stops after delivery: DNA and ALT every 1 to 3 months for 6 months



AASLD/IDSA 2025 R1 (PMID 41186418); CDC 2023 (PMID 36893044); CDC perinatal hepatitis B guidance.

Coinfection with HIV or HCV

Coinfection	What to do
HIV	ART must include TDF or TAF plus emtricitabine or lamivudine. Entecavir alone is not HIV treatment
HCV, HBsAg positive	DNA every 4 to 8 weeks during direct-acting antivirals and for 3 months after
HCV, HBsAg negative and anti-HBc positive	Much lower risk. Check DNA and HBsAg if the ALT rises unexpectedly

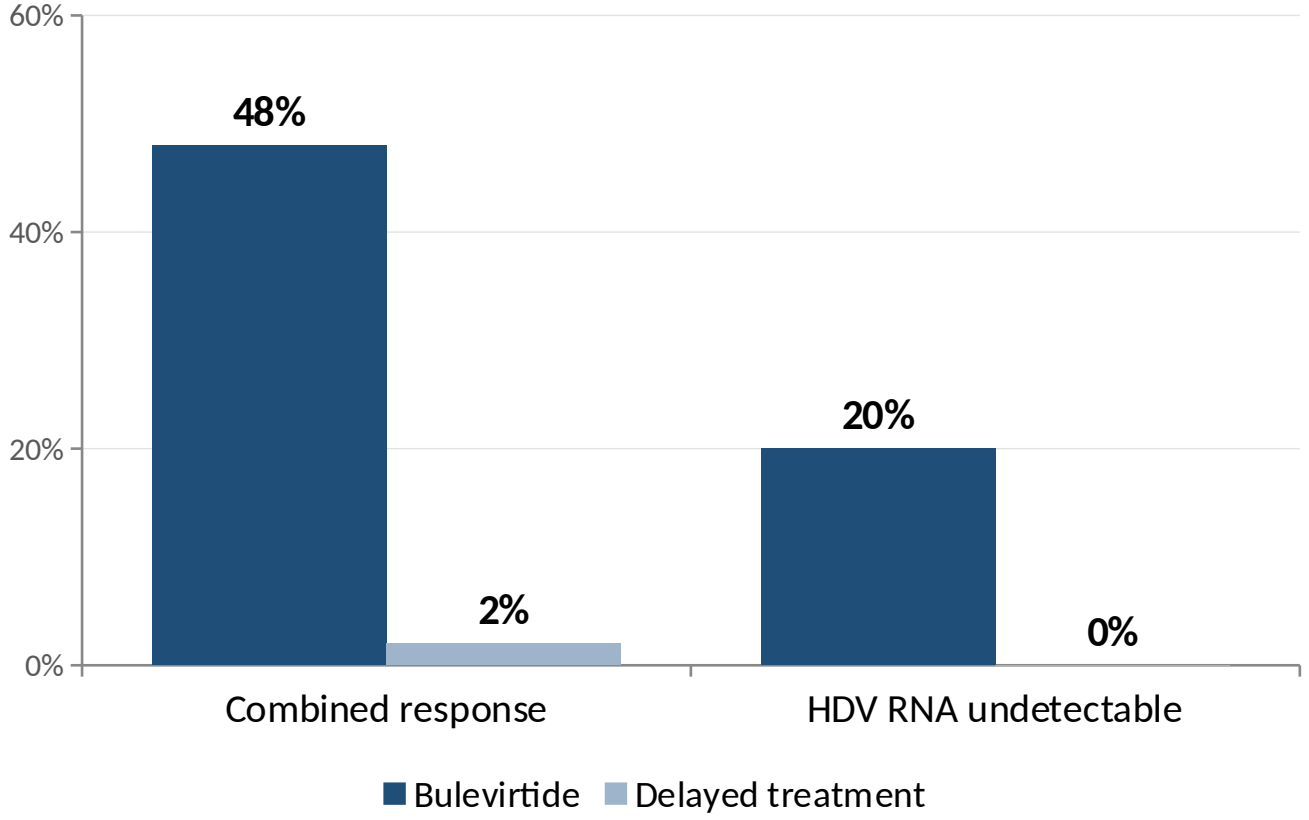
AASLD 2018, PMID 29405329.

Hepatitis D: who to test

	AASLD 2018	EASL 2025
Who	Risk-based: endemic origin, HIV, injection drug use, high ALT with low DNA	Every HBsAg-positive patient, at least once
How	Total anti-HDV, then HDV RNA if positive	Total anti-HDV, then HDV RNA

AASLD 2018, PMID 29405329; EASL 2025, PMID 40348683.

Bulevirtide for hepatitis D

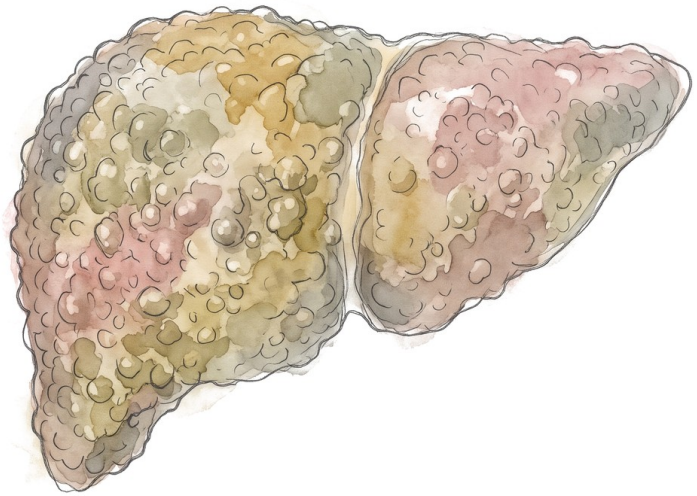


- FDA accelerated approval 22 May 2026
- 8.5 mg subcutaneously, once a day
- Adults without cirrhosis or with compensated cirrhosis
- Boxed warning: stopping can cause severe exacerbations

FDA news release 22 May 2026. MYR301, week 48. Combined response: HDV RNA undetectable or 2 log decline, and normal ALT.

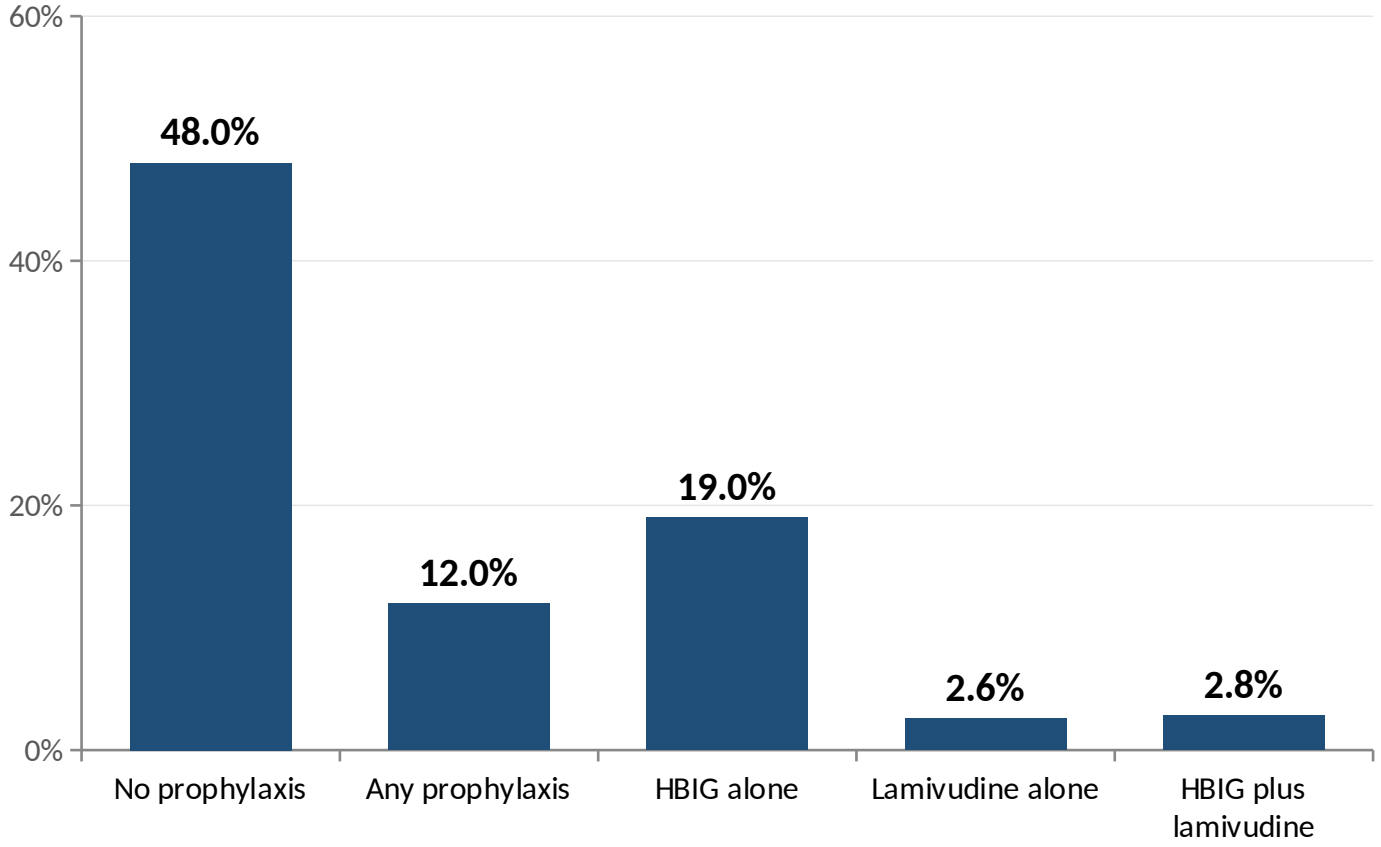
Cirrhosis and decompensation

Jang, observational	Treated	Untreated
First decompensation, n = 707	423	284
5-year transplant-free survival	59.7%	46.0%
Delisted on treatment	33.9%	



Jang JW et al. 2015 (dossier ref 56).

HBV and liver transplant



- Anti-HBc positive donor, HBV-naive recipient (Cholongitas, older regimens)
- Entecavir alone without HBIG: 265 recipients, 92% HBsAg negative and 100% DNA undetectable at 8 years (Fung)
- Long-term nucleos(t)ide analogue for HBsAg-negative recipients of anti-HBc-positive grafts

Cholongitas E et al. 2010 (dossier ref 59). Fung J et al. 2017 (ref 57). AASLD 2018 section 6H.

Children and adolescents

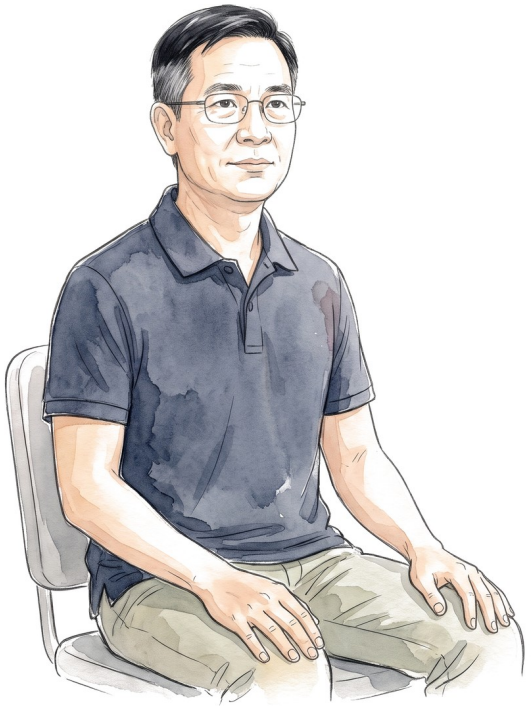
	Who is treated
AASLD 2018	HBeAg-positive, age 2 to under 18, raised ALT and measurable DNA. Normal ALT: no routine treatment
WHO 2024	Adolescents 12 and older use the adult pathways. Younger children are individualized
US labels	Entecavir and TDF from age 2. TAF from age 6 and 25 kg



AASLD 2018 R9A and R9B; WHO 2024 chapter 8; DailyMed labels.

Same man, 57 years old, five years on TAF

Condition for stopping	Our patient	Met?
No cirrhosis or advanced disease	Stiffness 5.9 kPa	
HBV DNA undetectable for 2 years or more	Undetectable for 5 years	
HBsAg under 100 IU/mL	400 IU/mL	
No HIV or HDV	Both negative	
Reliable follow-up	Yes	

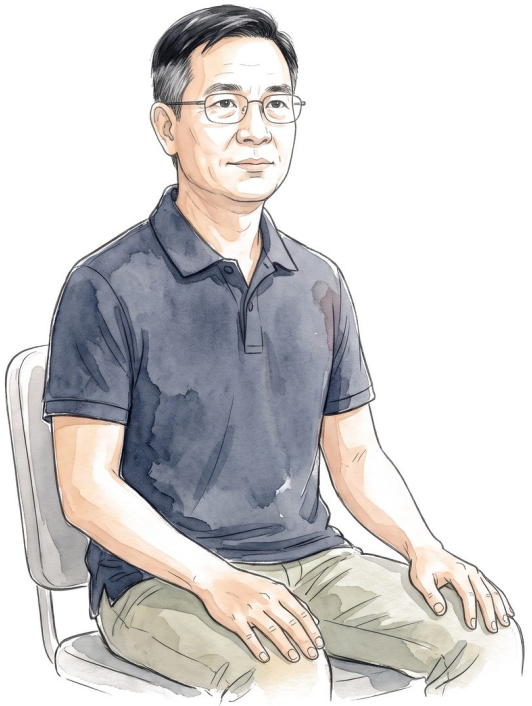


Can he stop?

AASLD/IDSA 2025 R5. Hepatology 2026;83:974-997, PMID 41186418.

Same man, 57 years old, five years on TAF

Condition for stopping	Our patient	Met?
No cirrhosis or advanced disease	Stiffness 5.9 kPa	Yes
HBV DNA undetectable for 2 years or more	Undetectable for 5 years	Yes
HBsAg under 100 IU/mL	400 IU/mL	**No**
No HIV or HDV	Both negative	Yes
Reliable follow-up	Yes	Yes



Can he stop?

No. Continue until HBsAg is lost

AASLD/IDSA 2025 R5. Hepatology 2026;83:974-997, PMID 41186418.

A different patient stops TAF

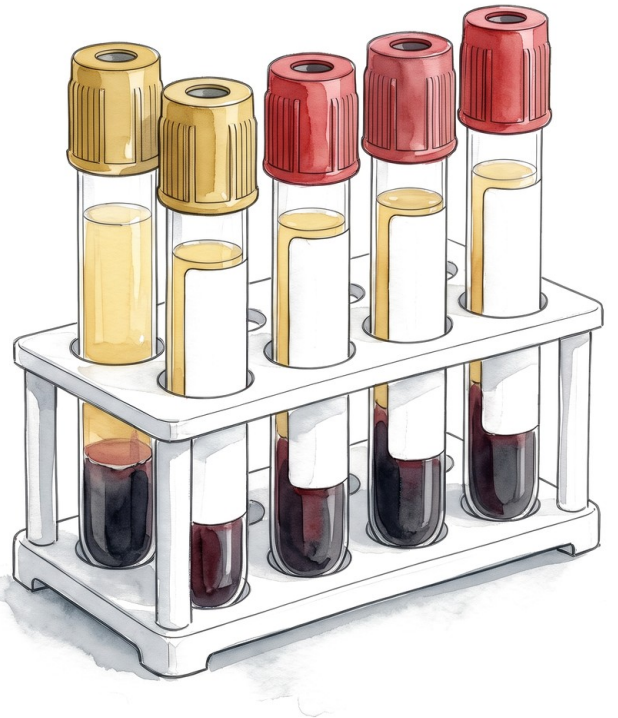
Which result means restart now?

A. ALT 1.5 times ULN

B. HBV DNA 10,000 IU/mL

C. HBsAg 200 IU/mL

D. Anti-HBs negative



AASLD/IDSA 2025 Table 3 (PMID 41186418).

A different patient stops TAF

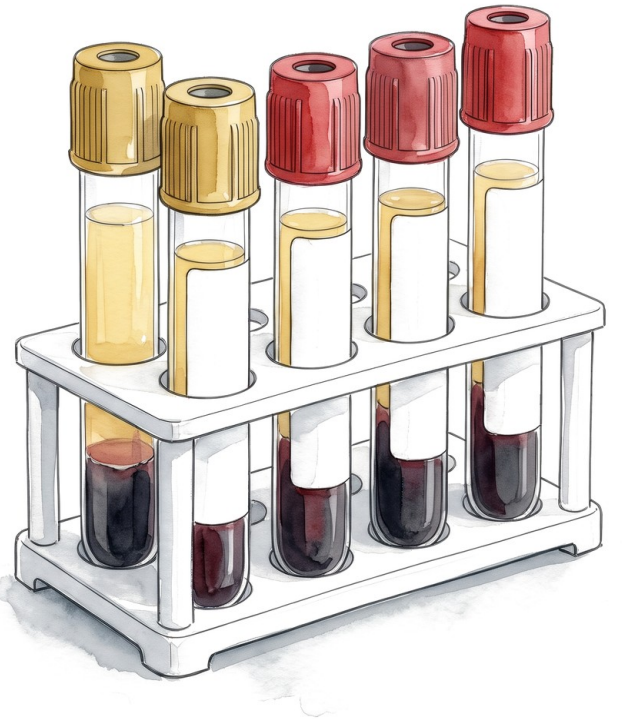
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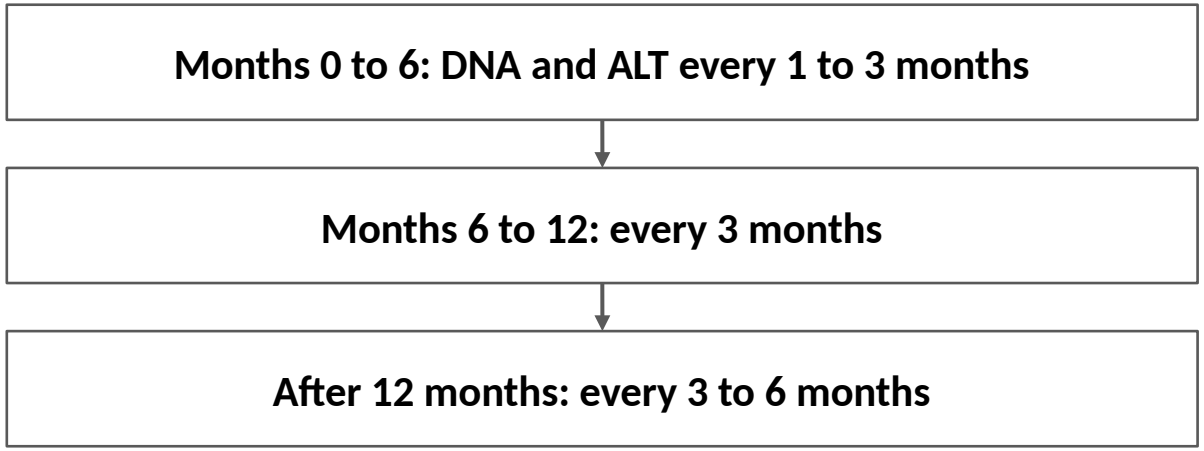


AASLD/IDSA 2025 Table 3 (PMID 41186418).

Stopping: summary

- **Default: continue until HBsAg is lost**
- **Stop only if all of these:**
- No cirrhosis
- DNA undetectable for 2 years or more
- HBsAg under 100 IU/mL
- No HIV or HDV
- Reliable follow-up

After stopping



Restart now: DNA 10,000 or more, ALT above 5 times ULN, bilirubin above 2.5 mg/dL, or decompensation

AASLD/IDSA 2025 R5 and Table 3 (PMID 41186418).

What happens when you stop

Study	Patients	HBsAg loss	Retreatment
STOP-NUC, randomized (Van Bommel 2023)	166	10.1% vs 0% at 96 weeks	13.9%
RETRACT-B, cohort (Hirode 2022)	1,552	3.2% at 12 months, 13.0% at 48 months	See paper
FINITE, randomized (Berg 2017)	42	4 of 21 (19%) in the stop arm at 144 weeks	13 of 21 stayed off therapy

Van Bommel F et al. J Hepatol 2023 (dossier ref 61); Hirode G et al. Gastroenterology 2022 (ref 60); Berg T et al. J Hepatol 2017 (ref 62).

Definitions of cure

Endpoint	Definition
Functional cure	HBsAg loss and HBV DNA below quantification, 24 weeks off therapy
Partial cure	HBsAg under 100 IU/mL and DNA below quantification, 24 weeks off therapy
Sterilizing cure	Elimination of the viral templates (cccDNA and integrated DNA)

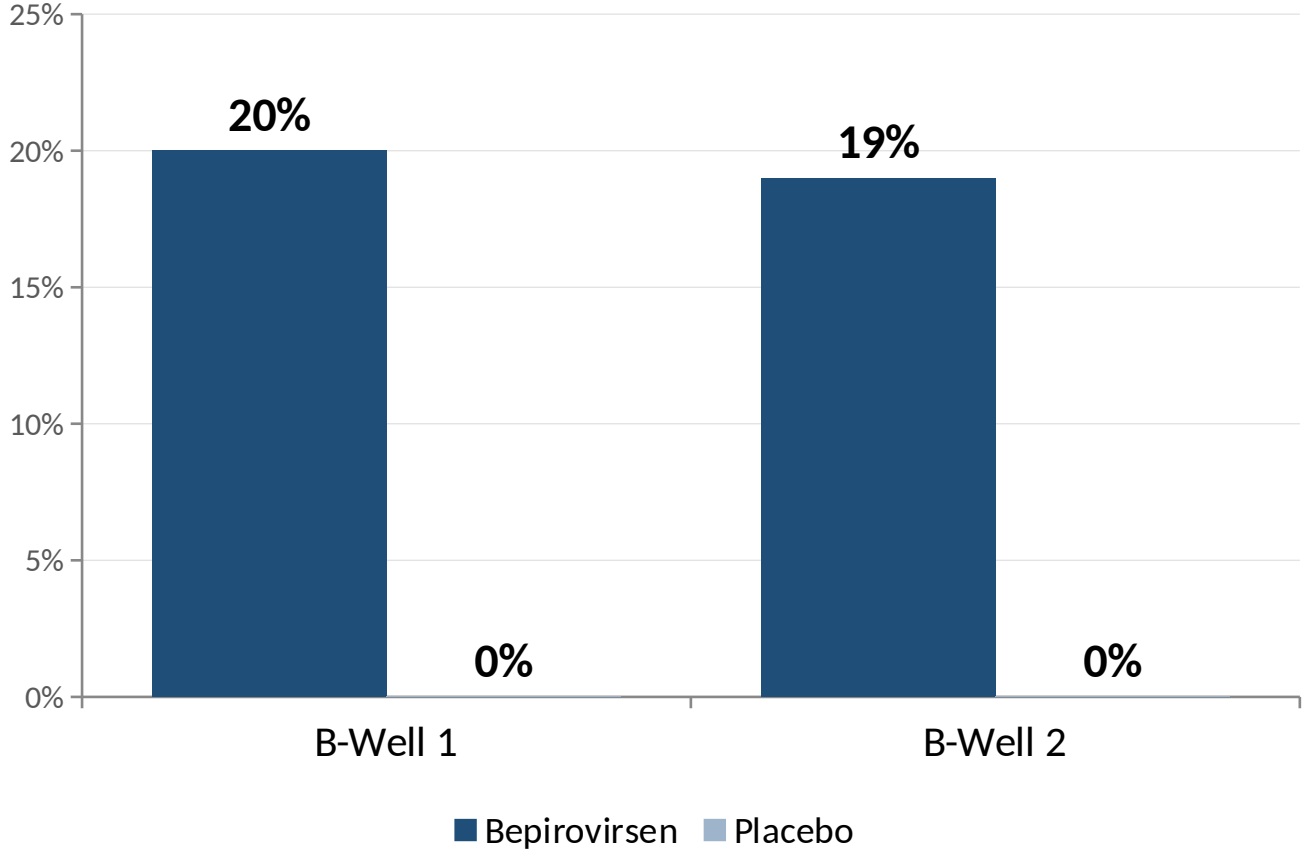
Ghany MG et al. 2023 (dossier ref 65).

Approved in 2026

	Bepirovirsen (Hibsago)	Bulevirtide (Hepcludex)
For	Chronic hepatitis B	Chronic hepatitis D
FDA approval	20 October 2026	22 May 2026
Who	Adults without cirrhosis, on a nucleos(t)ide analogue 6 months or more, DNA suppressed, HBsAg 3,000 IU/mL or lower	Adults without cirrhosis or with compensated cirrhosis
Result	19% had HBsAg undetectable 24 weeks after stopping all treatment; 0% on placebo	Week 48: combined response 48% vs 2%
Warning	Liver reactions, low platelets	Severe exacerbation if stopped

FDA news releases 20 October 2026 (bepirovirsen) and 22 May 2026 (bulevirtide). B-Well 1 and 2, Hou, NEJM 2026.

Bepirovirsen: B-Well 1 and 2



- Functional cure at week 72: HBsAg loss and DNA below quantification at least 24 weeks off all therapy
- 127 of 650 vs 0 of 328, and 106 of 570 vs 0 of 286
- Noncirrhotic, stable NA, HBsAg 100 to 3,000 IU/mL
- Grade 3 or higher adverse events 16% vs 3%

Hou J et al. N Engl J Med 2026 (B-Well 1 and 2, dossier ref 67). Final issue 25 June 2026.

In the pipeline

Agent	Phase 2 result	Caution
Xalnesiran (siRNA)	HBsAg loss at 24 weeks after treatment 7% (100 mg), 3% (200 mg), 12% with ruzotolimod, 23% with PegIFN, 0% NA alone	Background NA continued
Elebsiran with PegIFN	HBsAg loss 21% (200 mg) and 33% (100 mg) vs 6% with PegIFN alone	NA stopping follow-up ongoing
JNJ-3989 (siRNA)	Met NA-stopping criteria at week 48 in 5, 16 and 19% (40, 100, 200 mg) vs 2%	Stopping eligibility, not cure
Imdusiran (siRNA)	Phase 1a/b, safety primary	No cure rate yet

Dossier refs 66 to 71 and registry entries NCT04225715 and NCT06245291.

Back to our patients

	Patient	Decision
1	Man, 34 to 57. HBeAg-positive, then HBeAg-negative with DNA 8,000 and ALT 48	Monitor at 34. Treat at 41 (age) or at 52 (grey zone). PAGE-B 18, surveillance. Do not stop with HBsAg 400
2	Woman, 61. Anti-HBc positive, HBsAg negative, rituximab	Entecavir, TDF or TAF before the first dose and for 12 months after the last
3	Woman, 29, 20 weeks pregnant. DNA 5,000,000	TDF or TAF from week 28. Vaccine and HBIG within 12 hours. Test the baby at 9 to 12 months



Questions?

X: @mfalmeqdadi

Slides (PDF)